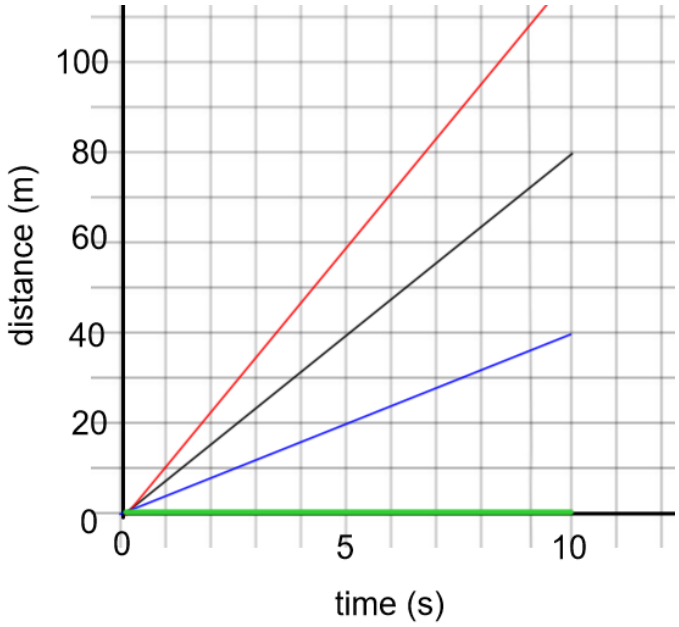
Year 10 Physics Workbook



<u>The Polesworth School</u>

Forces and Motion

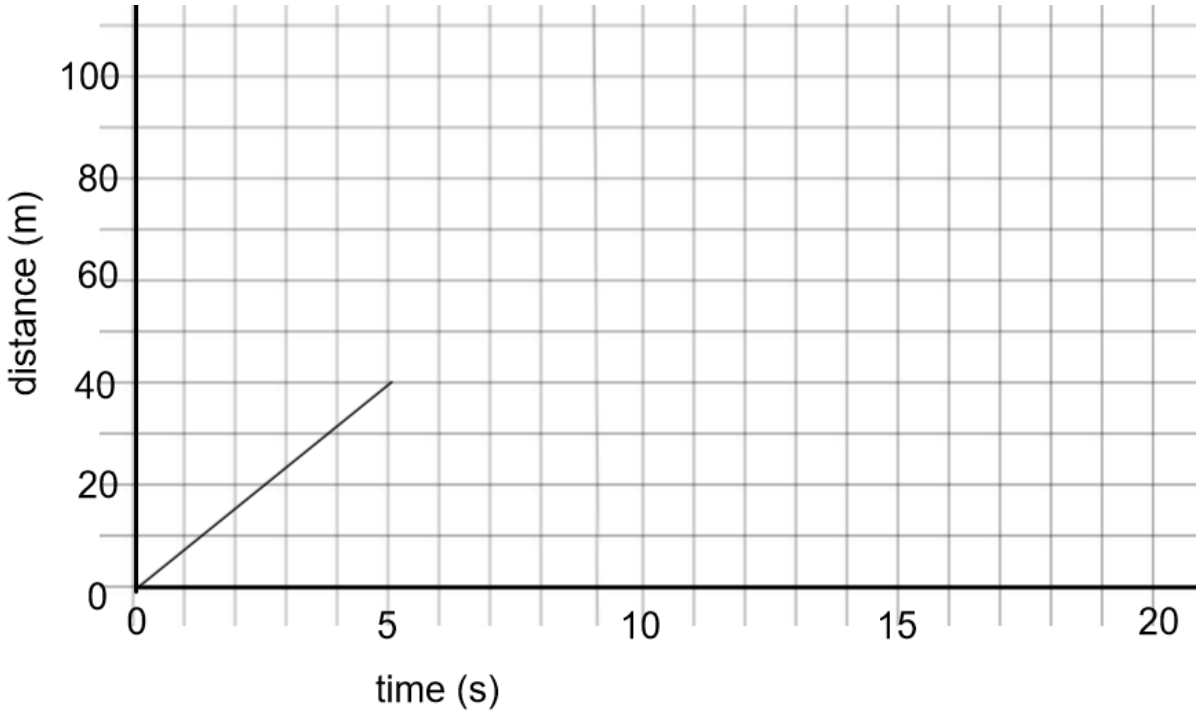
By the end of this booklet you should know the answers to all the common questions below

	Question	Answer
1	What is a "scalar quantity"?	A quantity with magnitude (size) only
2	What is a "vector quantity"?	A quantity with both magnitude and direction
3	Define "speed"	Rate of change of distance
4	Define "velocity"	Rate of change of displacement
5	Define "displacement"	Distance and direction
6	Give three examples of scalar quantities	Distance, mass, energy
7	Give five examples of vector quantities	Displacement, weight, force, velocity, acceleration
8	Define "acceleration"	Rate of change of velocity
9	What is a typical speed for walking?	1.5m/s
10	What is a typical speed for running?	3m/s
11	What is a typical speed for cycling?	6m/s
12	What is a typical speed for a car?	15m/s
13	What is a typical speed for a train?	90m/s
14	What is a typical speed for a plane?	250 m/s
15	What is a typical speed for sound in air?	330m/s
16	What formula relates speed, distance and time?	Speed = distance/ time
17	Describe speed and velocity in circular motion (HT)	Constant speed, changing velocity
18	Give the formula for acceleration	Acceleration = Change in velocity / time
19	What is acceleration due to gravity near Earth's surface?	$9.8m/s^2$

Distance-time graphs

If an object moves along a straight line, the distance travelled can be represented by a distance–time graph. The speed of an object can be calculated from the gradient of its distance–time graph.

These graphs always have time on the X-axis. We can think of them as telling a story: what happened first, second, etc.



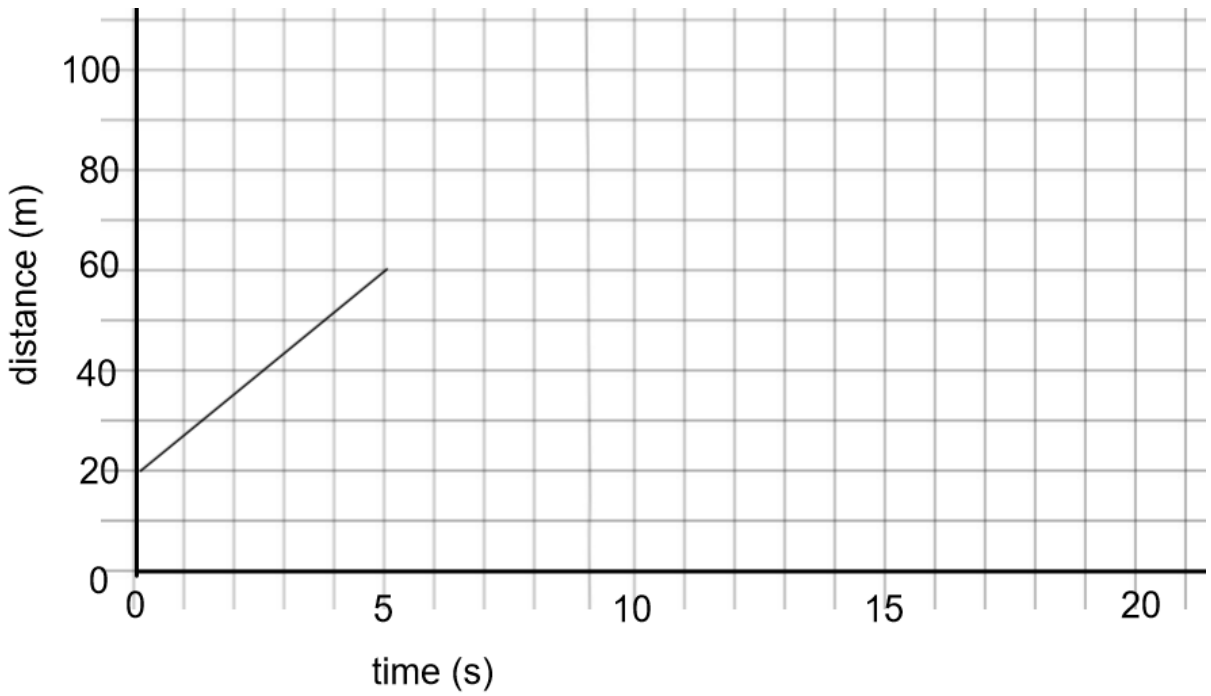
↑ This graph tells us that in 5 seconds an object travelled 40 metres.

Because speed = distance ÷ time, we can use values from the graph to calculate the speed:

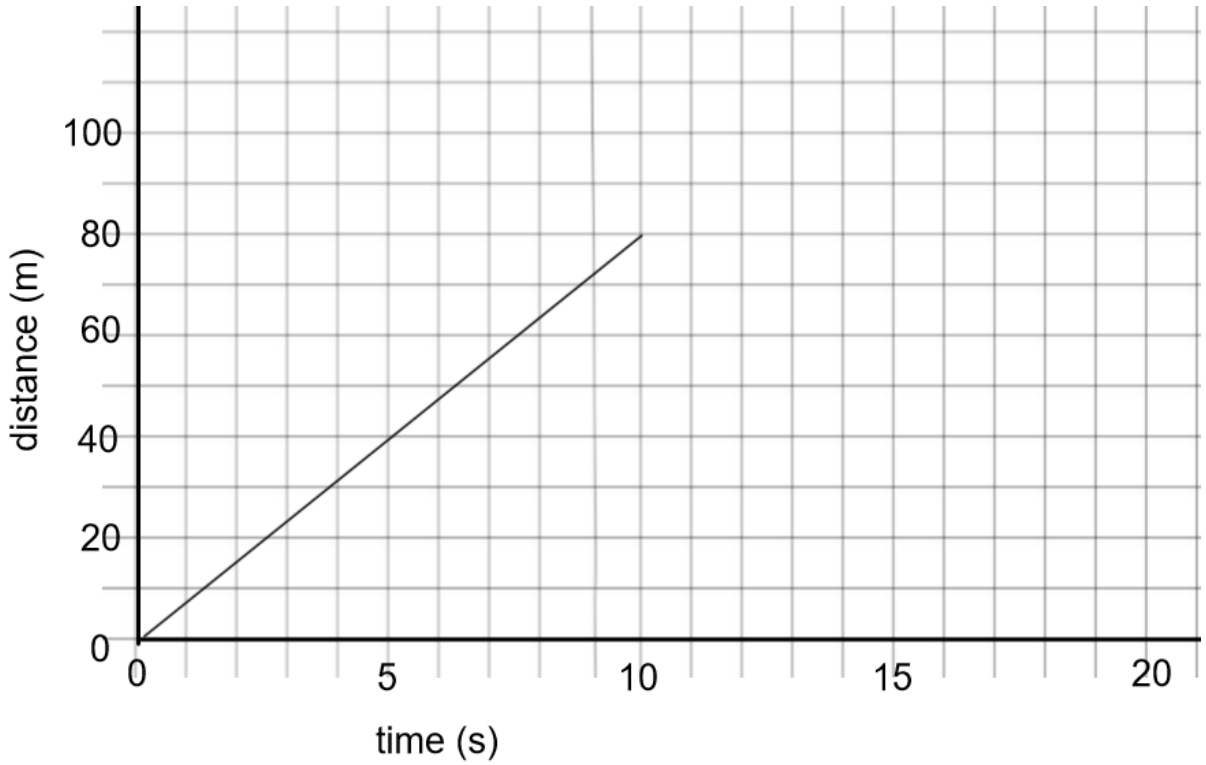
40 ÷ 5 = 8

Speed = 8m/s

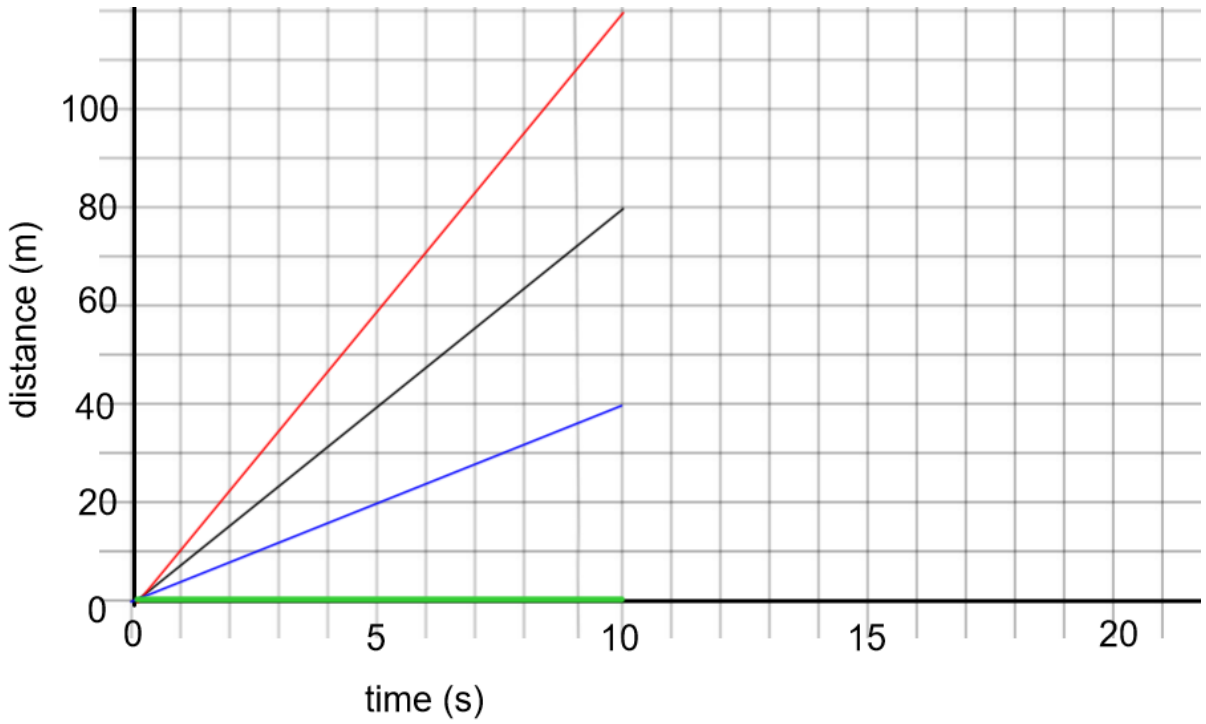
This is the same as the gradient: the gradient is the speed.



↑ This graph also shows us that an object travelled 40m in 5s. The gradient is the same. The difference is that the object started at 20m instead of 0.



↑ In this graph, the object travels 80m in 10s. The gradient is still 8 so the speed is still 8m/s.



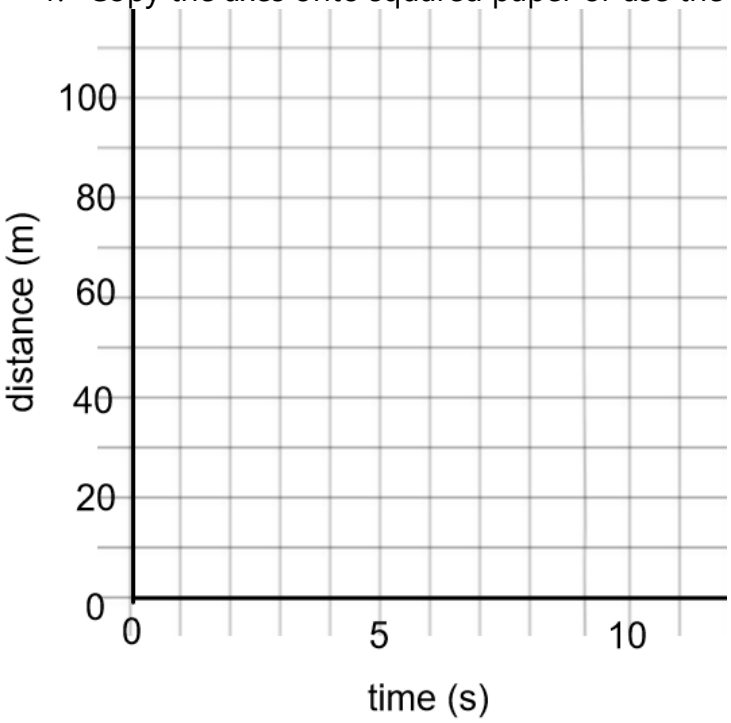
↑ A higher (steeper) gradient means a higher speed. In the graph above, the red line has a gradient of 120 ÷ 10 = 12m/s.

The blue line has a gradient of 40 ÷ 10 = 4m/s.

The green line has a gradient of 0, i.e. the object is stationary (stood still).

Questions

1. Copy the axes onto squared paper or use the one below:

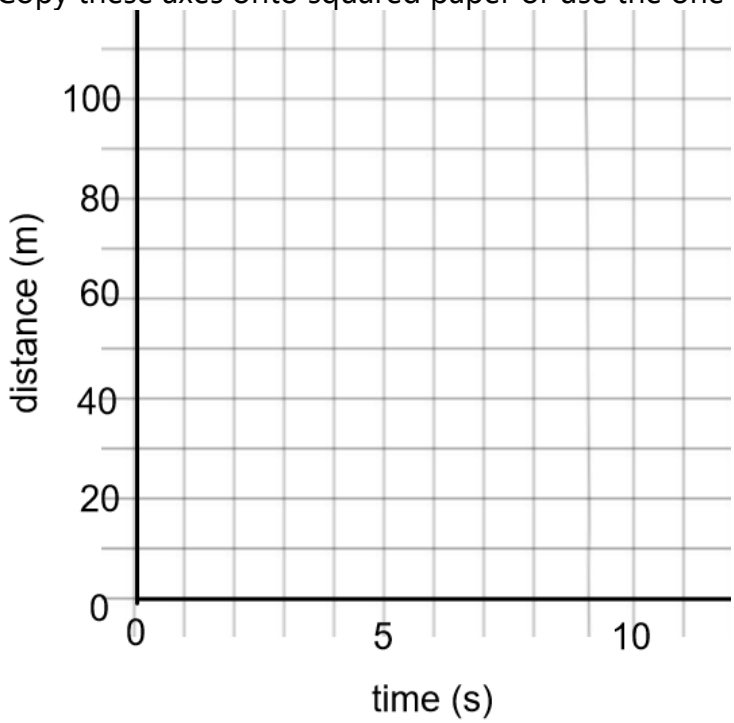


Draw lines to show the motion of an object that travels:

a) 60m in 5s
b) 100m in 5s
c) 20m in 5s

2. Find the speed for each of the objects in (1)

3. Copy these axes onto squared paper or use the one below:



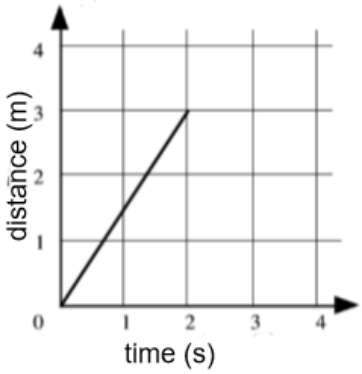
Draw lines to show the motion of:

a) An object that travels 10m in 10s
b) An object that travels 100m in 10s
c) An object that travels 40m in 10s.

4. Find the speeds of each of these objects.

5. Find the speed of the following:

a)



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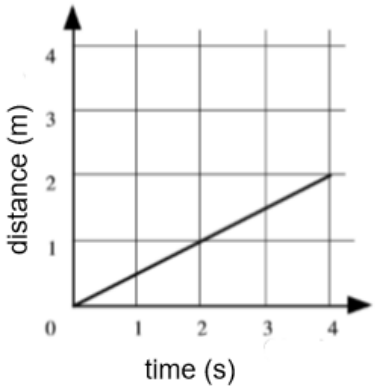

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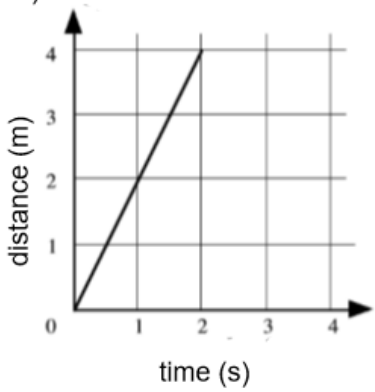

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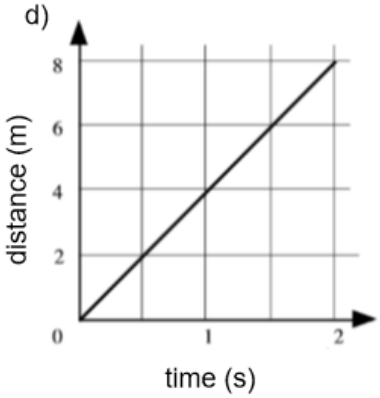

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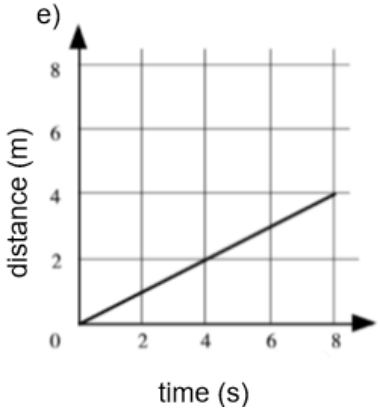

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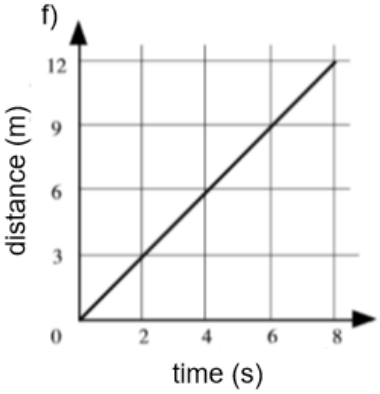

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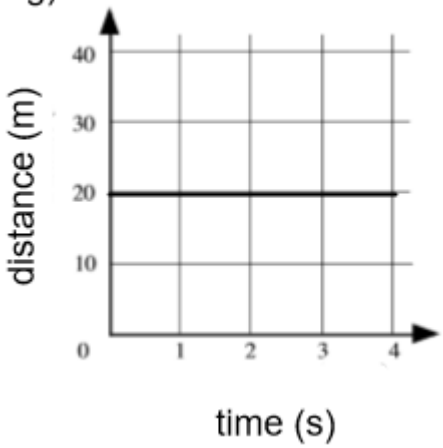
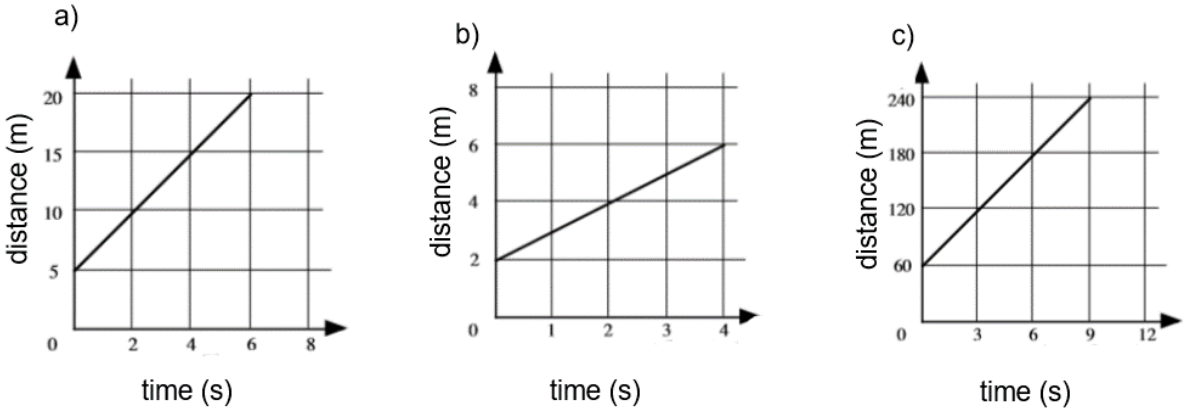
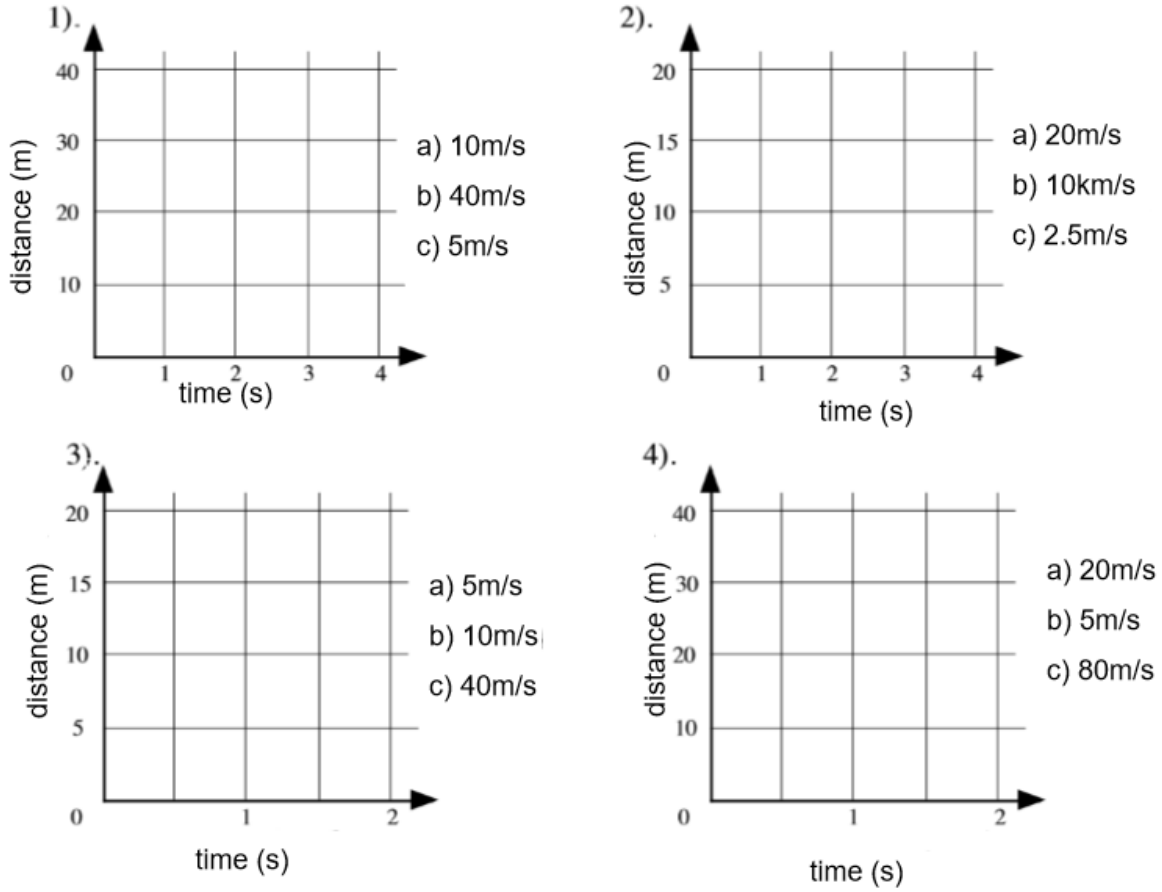


6. Find the speed of the following:



Extension

For each question, copy the axes onto squared paper, then draw a line to show an object moving at each of the three speeds given.

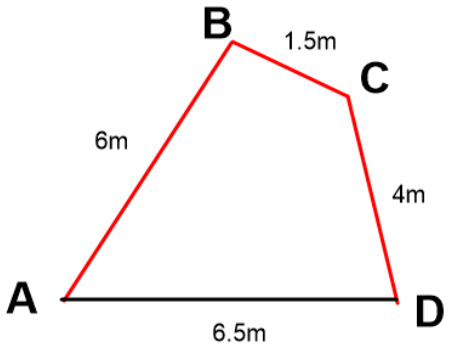


Distance and displacement

An object has a **position** relative to its surroundings at any instant in time. When an object moves, it changes its position.

Distance is a **scalar** quantity that refers to "how much ground an object has covered" during its motion. Scalar quantities just have a number (magnitude).

Displacement is a **vector** quantity that refers to "how far and in what direction the object's position has changed". Vector quantities have a magnitude and direction.

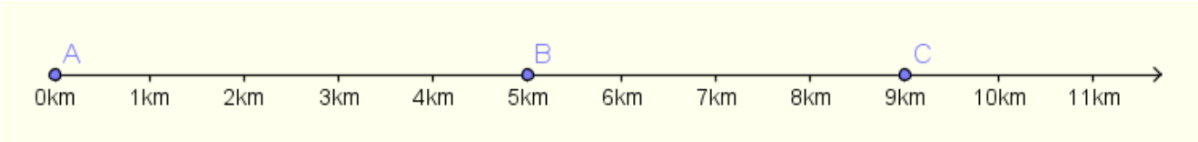


Consider an object that starts at position A, and then moves to B, C and finally to D.

The distance travelled by the object is (6 + 1.5 + 4) = 11.5m.

The displacement of the object is 6.5m to the right.

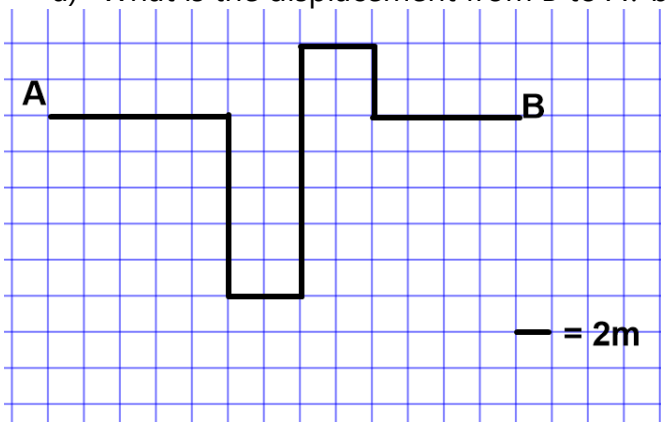
Questions

1. Define "distance"
2. Define "displacement".
3. Draw a diagram to show the difference between distance and displacement.
4. Which quantity is a vector quantity: distance or displacement?
5. When people say "As the crow flies" are they referring to distance or displacement?
6. True or false: For a single object in motion, displacement can be equal to distance but it can never be bigger.
7. 

   A car starts at A and travels to B, then to C, and then back to B. What is a) the distance travelled and b) the displacement of the car?

8.

a) What is the displacement from B to A? b) What is the distance?



9.

A walker travels from A to B, C, D, E and F. What is the distance travelled? What is the displacement?

Calculating velocity and speed

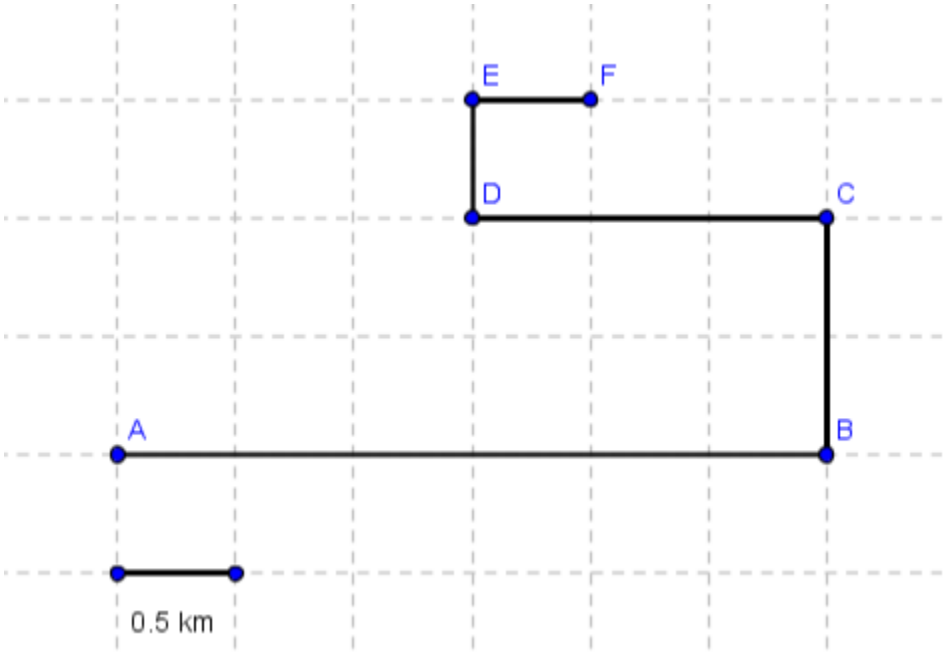
Speed is the rate of change of distance:

Speed = distance ÷ time
Distance = speed x time

Speed is a scalar quantity: it only has magnitude, e.g. 14m/s

Velocity is like speed, but it's a vector quantity, so it always has direction as well as speed. This is because velocity is rate of change of displacement and displacement is a vector quantity.

Velocity = displacement ÷ time



Displacement = velocity x time

However, people often use the word “velocity” without stating a direction – this is sloppy but very common.

Questions

1. Define speed
2. Define velocity
3. What is the difference between speed and velocity?
4. Find the velocity of a car which travels 25m to the right over a period of 10 seconds.
5. What is the velocity of a hot air balloon which travels 125m up in a time of 50 seconds?
6. A diver travels 640m down during a 16-second dive. Find his velocity.
7. What is the displacement of a rocket travelling at a velocity of 150m/s up for 30 seconds?
8. Find the displacement of a beetle crawling for 180 seconds at a velocity of 0.25m/s to the left.
9. A car has a velocity of 24m/s to the right and travels for 58 seconds. Find its displacement.
10. A car travelling at a velocity of 2.75m/s travels for 185 meters. What time was taken?
11. Find the time taken by a rocket travelling at a velocity of 240m/s to reach a displacement of 15,000m.
12. A snail travels with a velocity of 0.08m/s. It reaches a displacement of 0.2m. Find the time taken.

Acceleration

In everyday language, "accelerate" just means "go faster". In Physics, it means more:

Acceleration is defined as "rate of change of velocity (speed OR direction)".

Since velocity is both **speed and direction**, several different things count as acceleration:

- Speeding up (increasing speed)
- Starting to move from rest (increasing speed from 0)
- Changing direction of linear motion e.g. a ball bouncing (changing speed from + to – or vice versa)
- Slowing down (decreasing speed) – also called "deceleration"
- Stopping (decreasing speed to 0) – also called "deceleration"
- Turning a corner (changing direction)
- Circular motion (changing direction)

An acceleration always requires a **resultant force**. This is why you feel a force on you when you're in a car and it speeds up, slows down or turns a corner.

The units for acceleration are m/s^2.

Questions

1. Define "acceleration".
2. What is the unit for acceleration?
3. Explain why pressing the accelerator in a car causes the car to accelerate: what change is made to the velocity?
4. Explain why applying the brakes in a moving car causes it to accelerate: what change is made to the velocity?
5. Explain why the motion of a pebble dropped from a resting position is acceleration: what change is made to the velocity?
6. Explain why a pebble landing on the ground undergoes acceleration: what change is made to the velocity?
7. Explain why a bouncing ball is accelerating: what change is made to the velocity?
8. Explain why a car turning a corner is accelerating: what change is made to the velocity?
9. Explain why the Moon orbiting the Earth is accelerating.
10. What is the relationship between force and acceleration?

Acceleration formula 1

Some typical accelerations:
Motorbike: $6m/s^2$
Car: $3m/s^2$

Object falling on Earth: $9.8m/s^2$
Space shuttle launch: $29m/s^2$

Average acceleration (or deceleration) can be calculated using the formula:

Acceleration = change in velocity ÷ change in time

And rearranged:

Change in velocity = acceleration x time

↑Which you need to memorise.

The change in velocity can be found using:

Final velocity – initial velocity

If an object is slowing down, the change in velocity will be a negative number. This gives a negative value for the acceleration. A negative acceleration is also called "deceleration".

Questions

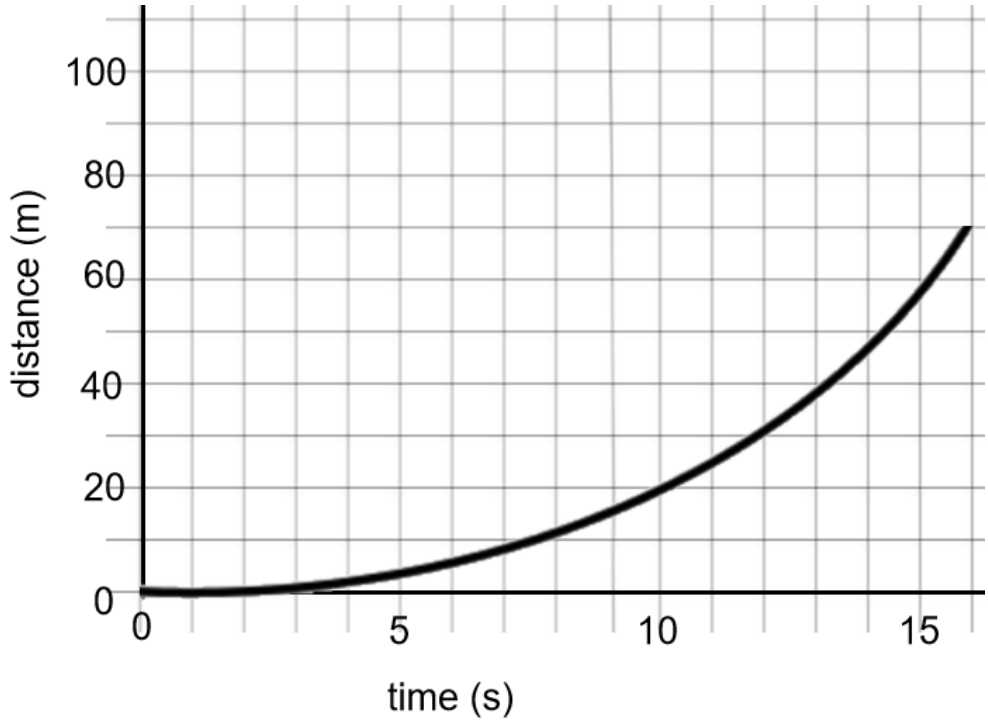
1. Give typical accelerations for a car, motorbike, object falling on Earth and a space shuttle launch.
2. What formula links change in speed, acceleration and change in time?

3. An object accelerates from 5m/s to 10m/s over a period of 20s. Find the acceleration.
4. Find the acceleration of an object which starts at 0.2m/s and increases its speed to 12.5m/s over a period of 5s.

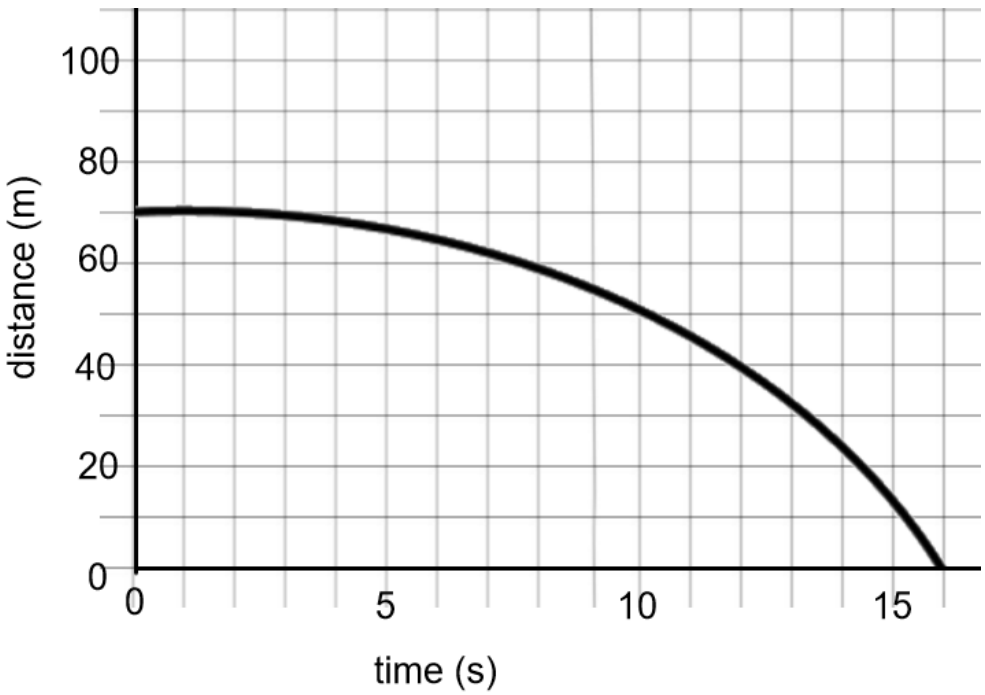
5. What acceleration takes an object to 8m/s from 20m/s over a period of 3s?
6. An object starts at 100m/s and decelerates to 75m/s over a period of 5 seconds. Calculate the deceleration.
7. Find the deceleration of an object starting out at 45m/s and reducing its speed to 13m/s over a period of 12s.
8. An object with initial velocity of 48m/s decelerates to 3m/s over a period of 30s. What is the deceleration?
9. An object starts at rest and accelerates to 14m/s over a period of 3.5s. Find the acceleration.
10. An object accelerates to 105m/s from rest in 55s. Calculate the acceleration.
11. What acceleration takes an object from rest to 64m/s in 4s?
12. An object travelling at 35m/s slows to a stop over a period of 70s. What is the deceleration?
13. Find the deceleration when an object comes to rest from 12m/s over a period of 2.5s.
14. What deceleration occurs when an object travelling at 58m/s comes to rest over a period of 4s?
15. How long would it take an object with deceleration $2m/s^2$ to go from 10m/s to 4m/s?
16. What time would it take an object with acceleration $4.3m/s^2$ to accelerate from 9m/s to 85m/s?
17. For an object with acceleration 0.45m/s, find the time taken to accelerate from rest to 3m/s.
18. For an object with an acceleration of $6m/s^2$, what is the change in velocity over 15s?
19. For an object with an acceleration of $-18m/s^2$, what is the change in velocity over 76s?
20. For an object with an acceleration of $0.9m/s^2$, what is the change in velocity over 200s?

Acceleration on a distance-time graph (HT)

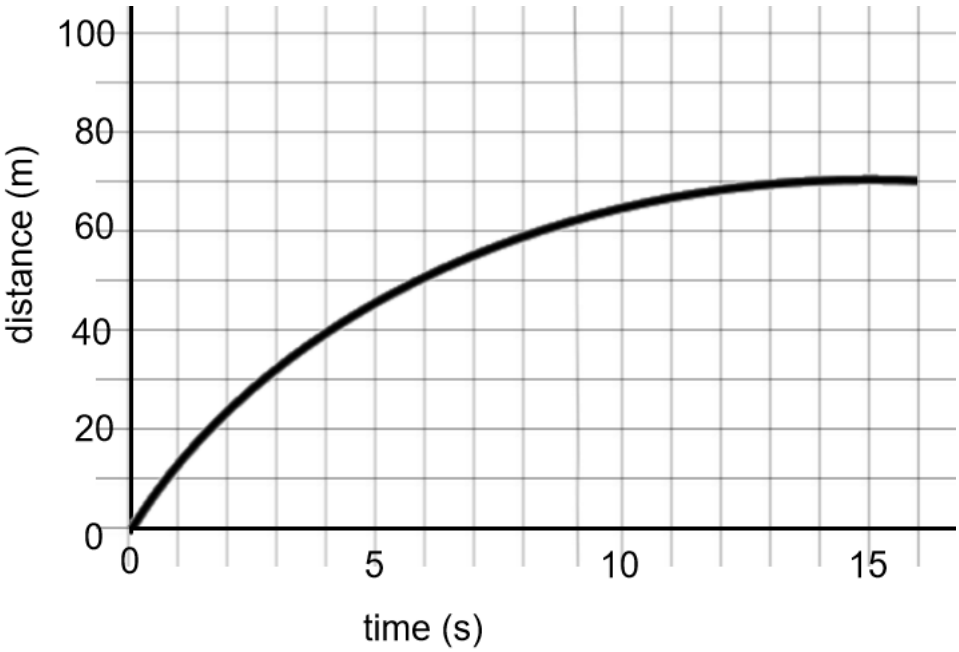
Acceleration is defined as rate of change of velocity. Since velocity is equivalent to the gradient on a distance-time graph, and accelerating object will have a changing gradient, i.e. a curved line:



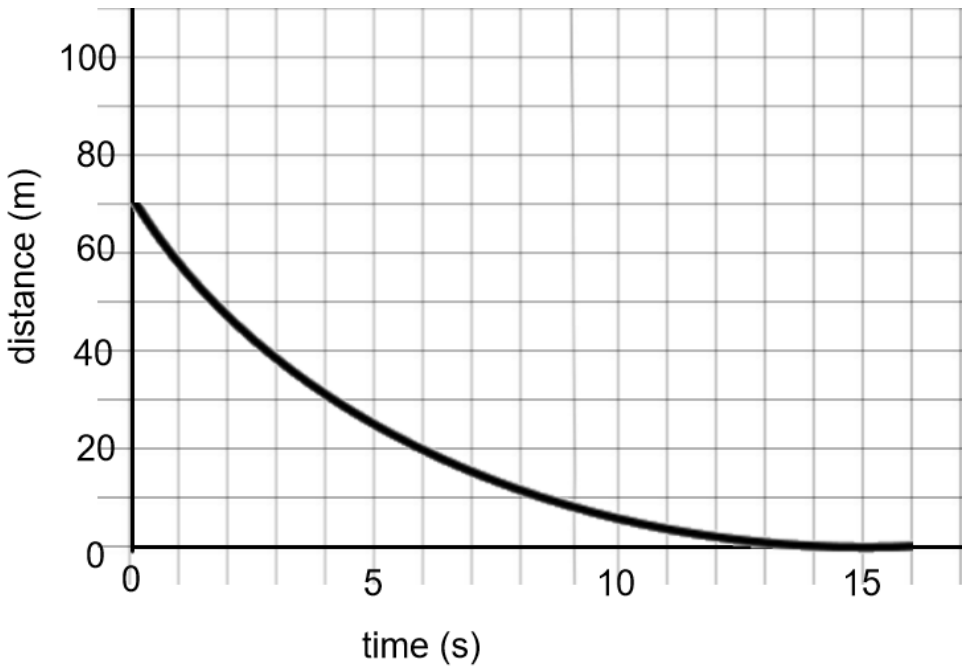
↑ This graph shows an object with increasing speed (positive acceleration) as the gradient is getting higher (steeper).



↑ This graph also shows an object with positive acceleration (increasing speed). It looks different because this object is travelling backwards instead of forwards.



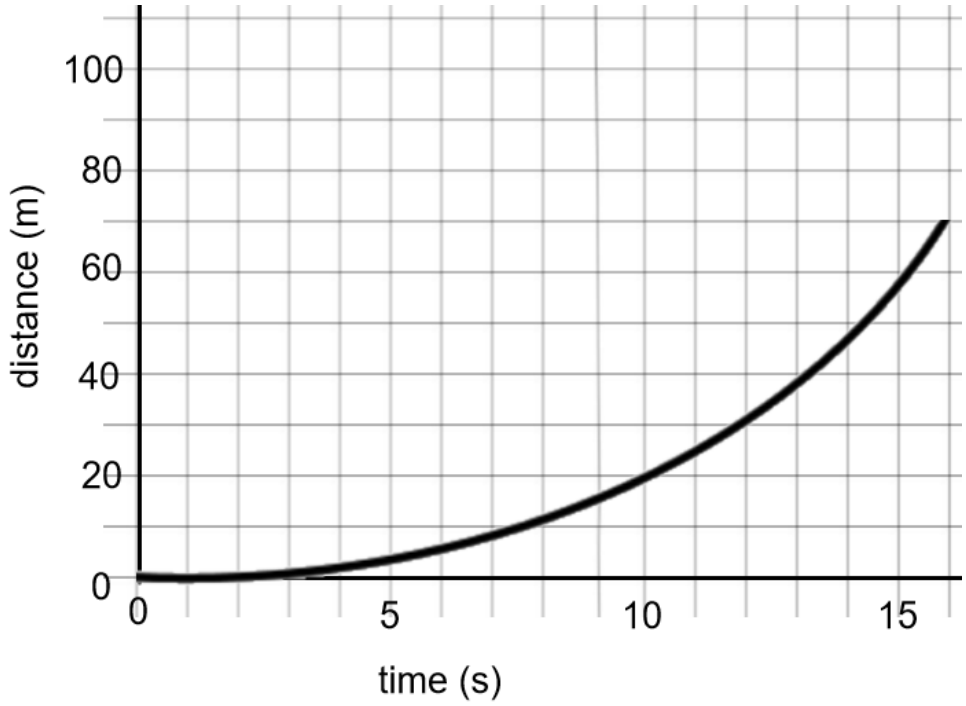
↑ This graph shows an object moving forward with negative acceleration, i.e. slowing down. The gradient is decreasing.

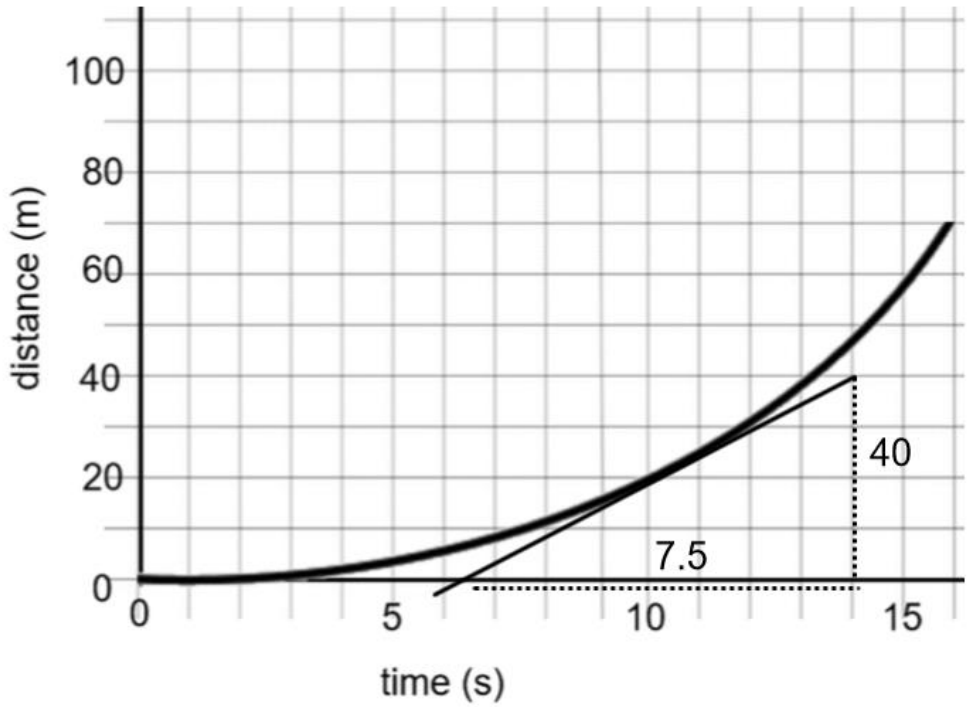


↑ This graph shows an object moving backwards with negative acceleration (slowing down).

To find the speed at any time, we can draw a tangent to the curve at that time, and find the gradient:

Example: Find the speed at 10 seconds:



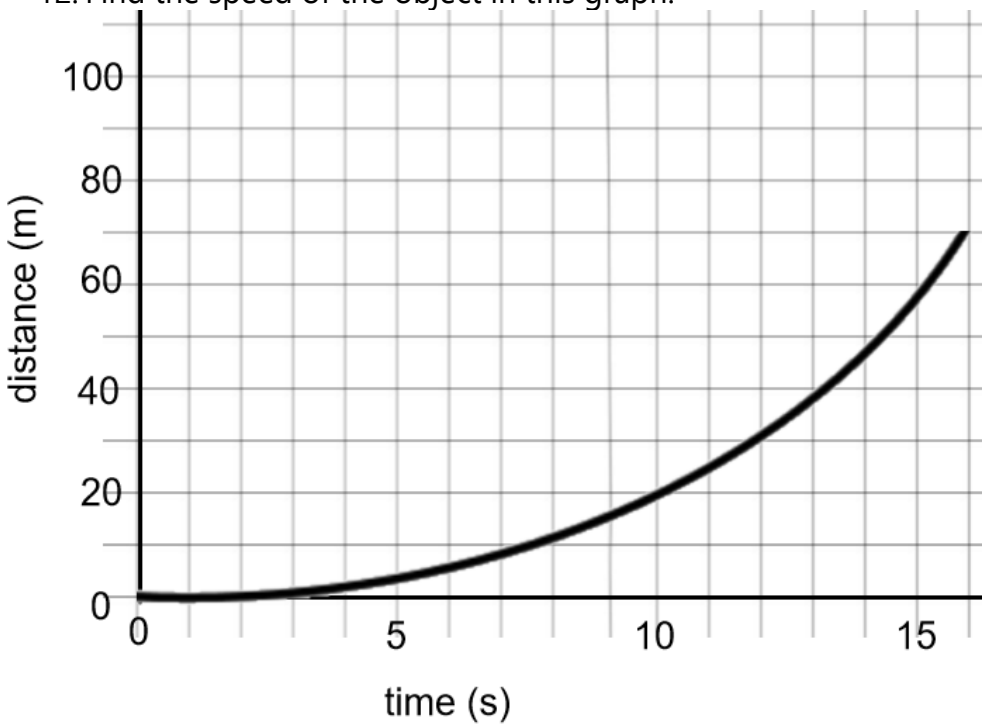


40 ÷ 7.5 = 5.33m/s

Questions:

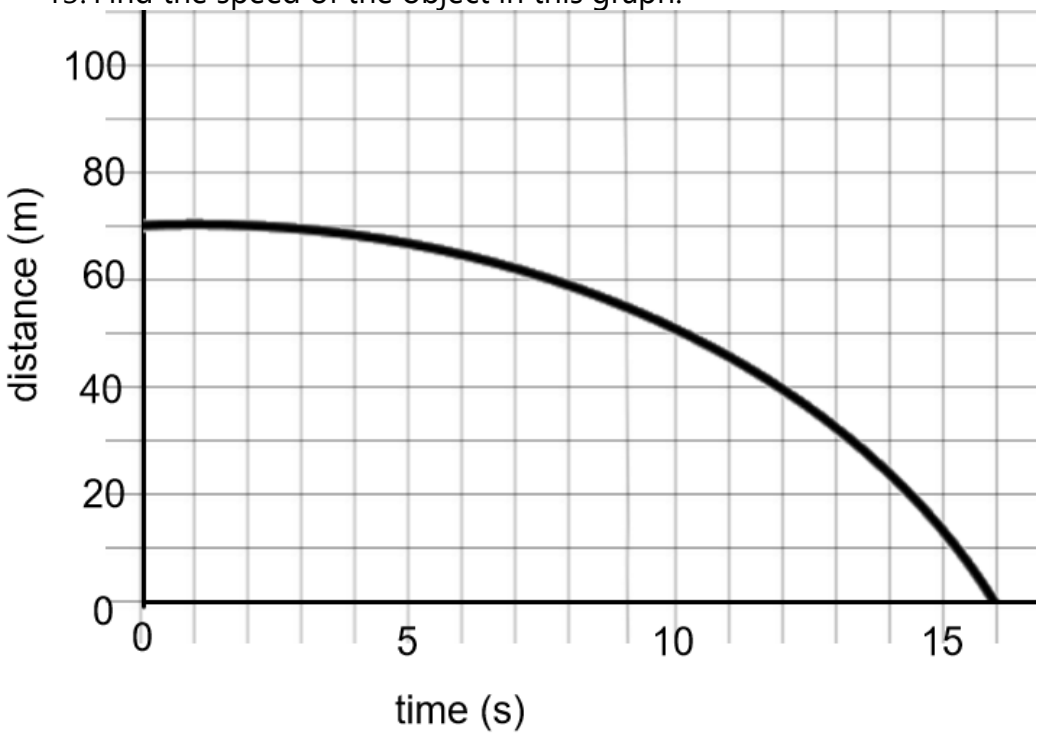
1. On a distance-time graph, what sort of line shows acceleration?
2. Sketch a d-t graph showing positive acceleration moving forwards
3. Sketch a d-t graph showing negative acceleration moving forwards
4. Sketch a d-t graph showing positive acceleration moving backwards
5. Sketch a d-t graph showing negative acceleration moving backwards
6. What happens to speed in positive acceleration?
7. What type of motion gives a positive gradient on a d-t graph?
8. What happens to speed in negative acceleration?
9. What type of motion gives a negative gradient on a d-t graph?
10. What is “negative acceleration” also known as?
11. How can we find momentary speed of an accelerating object from a distance-time graph?

12. Find the speed of the object in this graph:



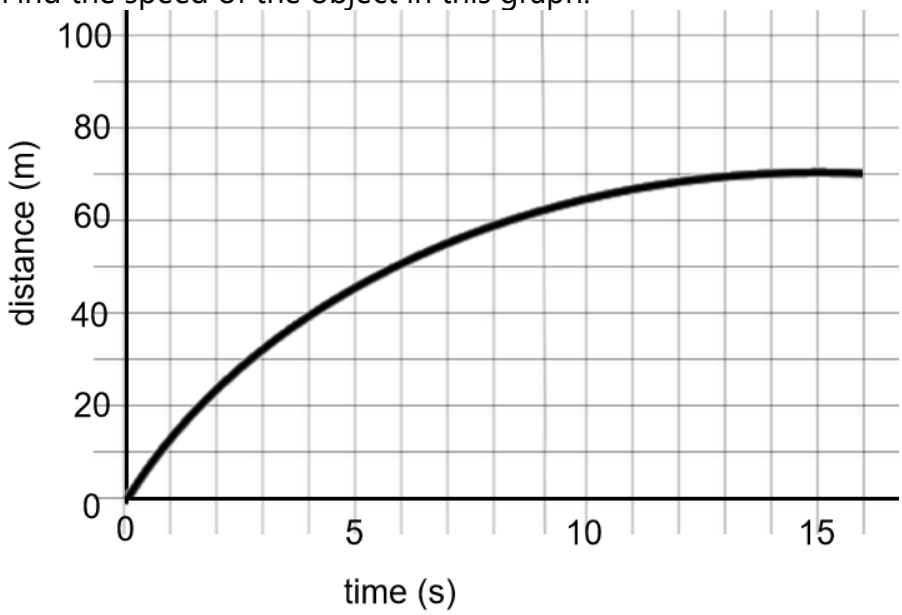
a. At 7 seconds
b. At 11 seconds
c. At 13 seconds
d. Is the object moving forwards or backwards?
e. Is the object accelerating or decelerating?

13. Find the speed of the object in this graph:



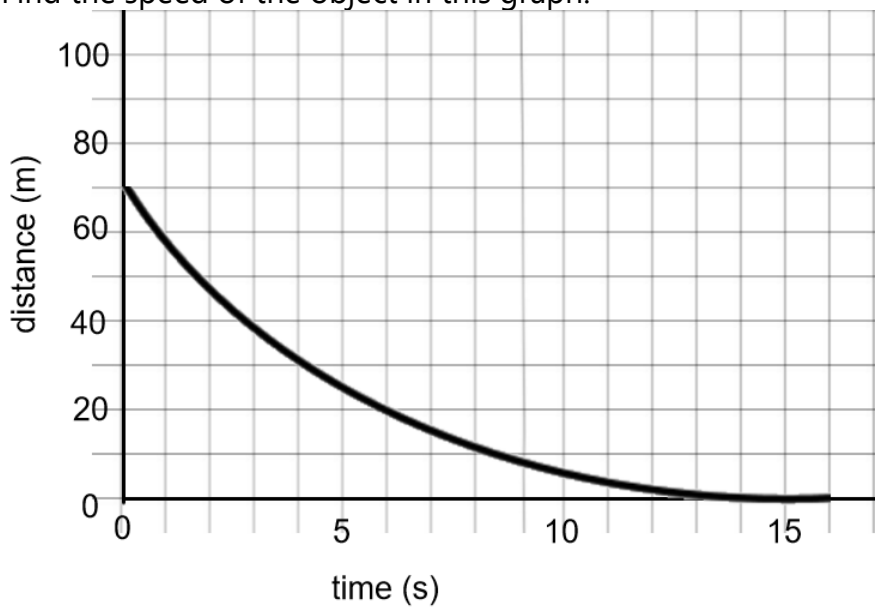
a. At 7 seconds
b. 10 seconds
c. 12 seconds
d. 14 seconds
e. Is the object moving forwards or backwards?
f. Is the object accelerating or decelerating?

14. Find the speed of the object in this graph:



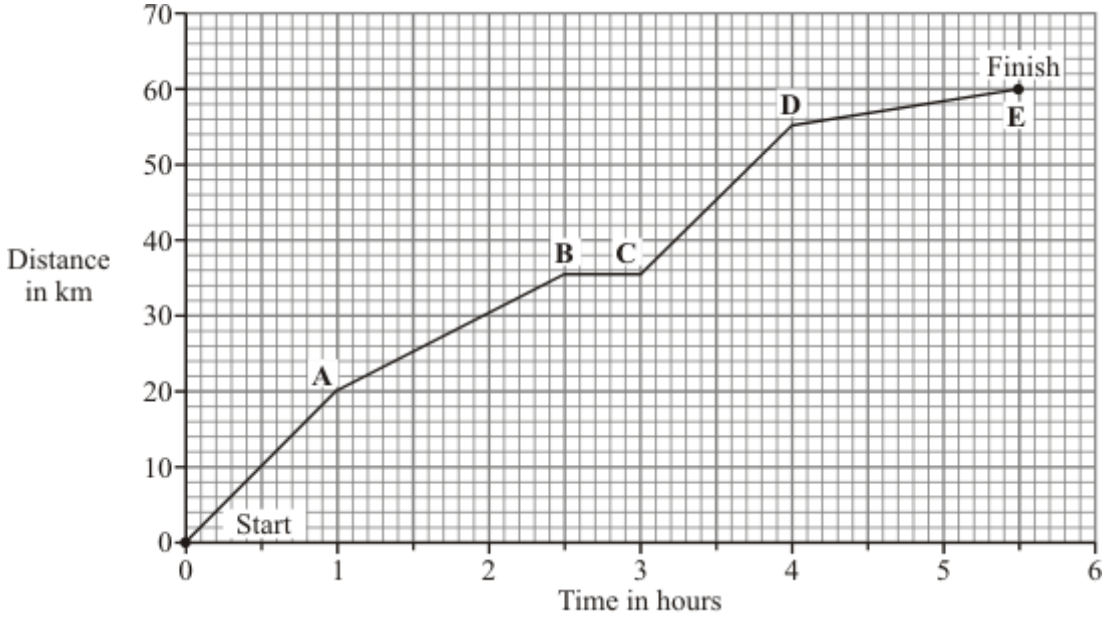
a. At 3 seconds
b. At 6 seconds
c. At 9 seconds
d. Is the object moving forwards or backwards?
e. Is the object accelerating or decelerating?

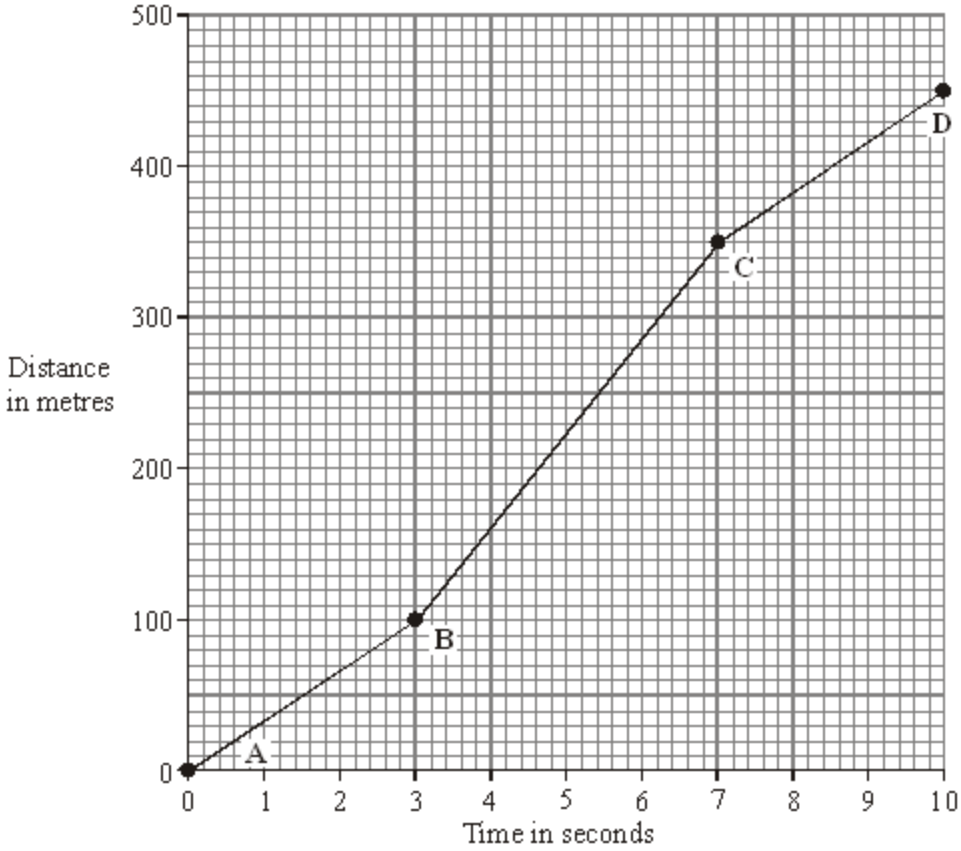
15. Find the speed of the object in this graph:

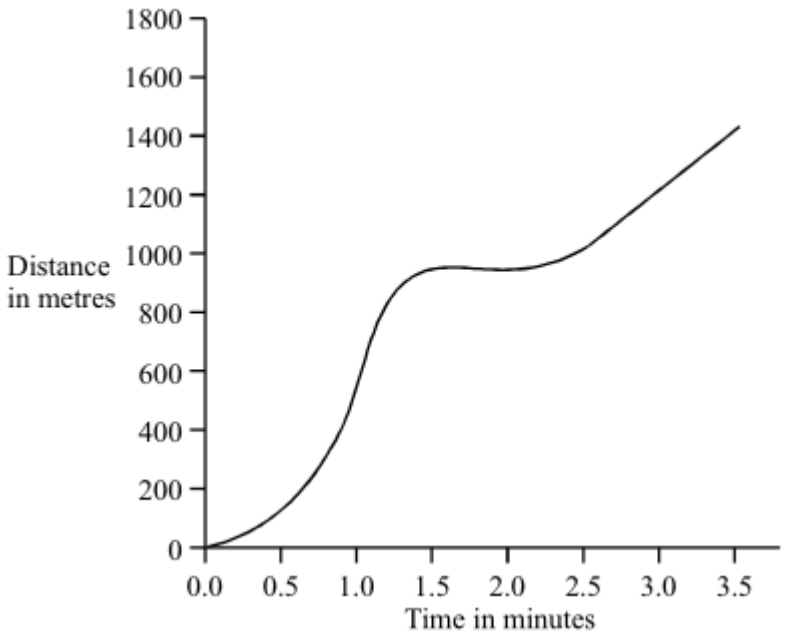


a. At 3 seconds
b. At 5 seconds
c. At 8 seconds.
d. Is the object moving forwards or backwards?
e. Is the object accelerating or decelerating?

Distance – time graphs goal-free problems

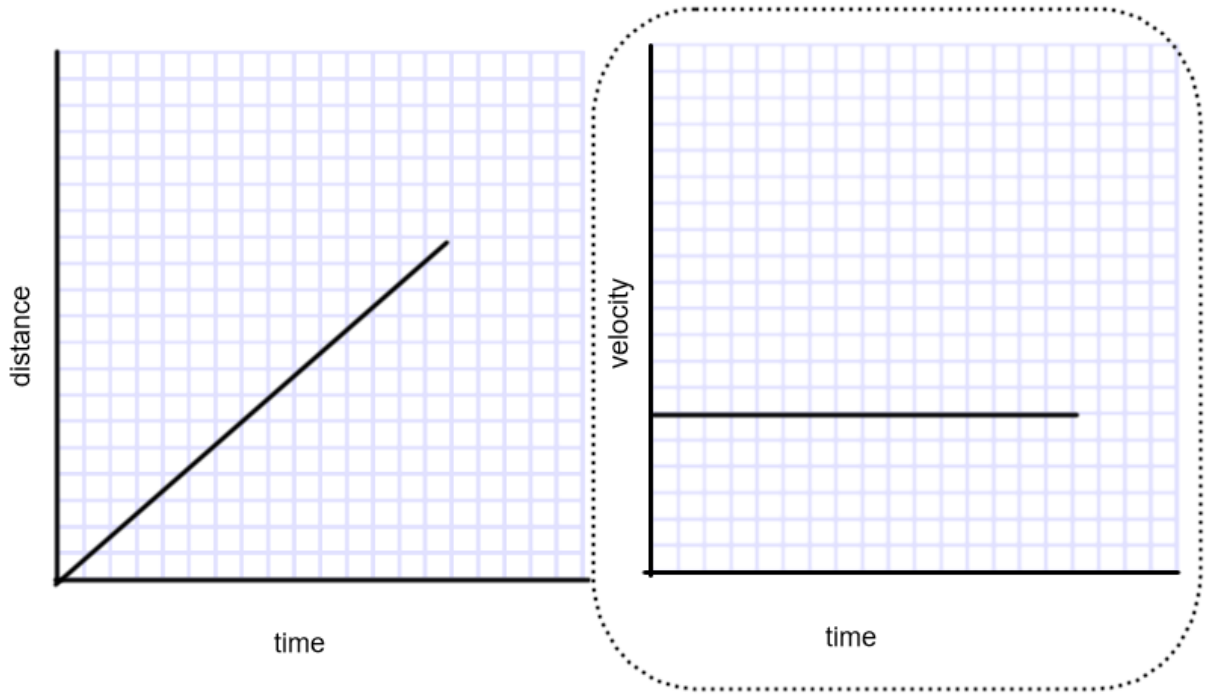




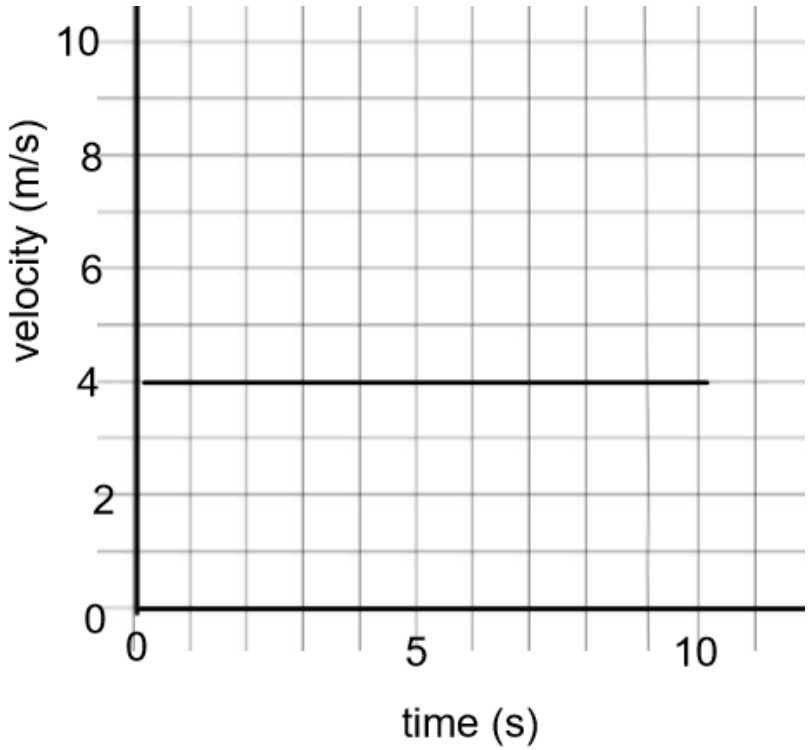


Velocity time graphs

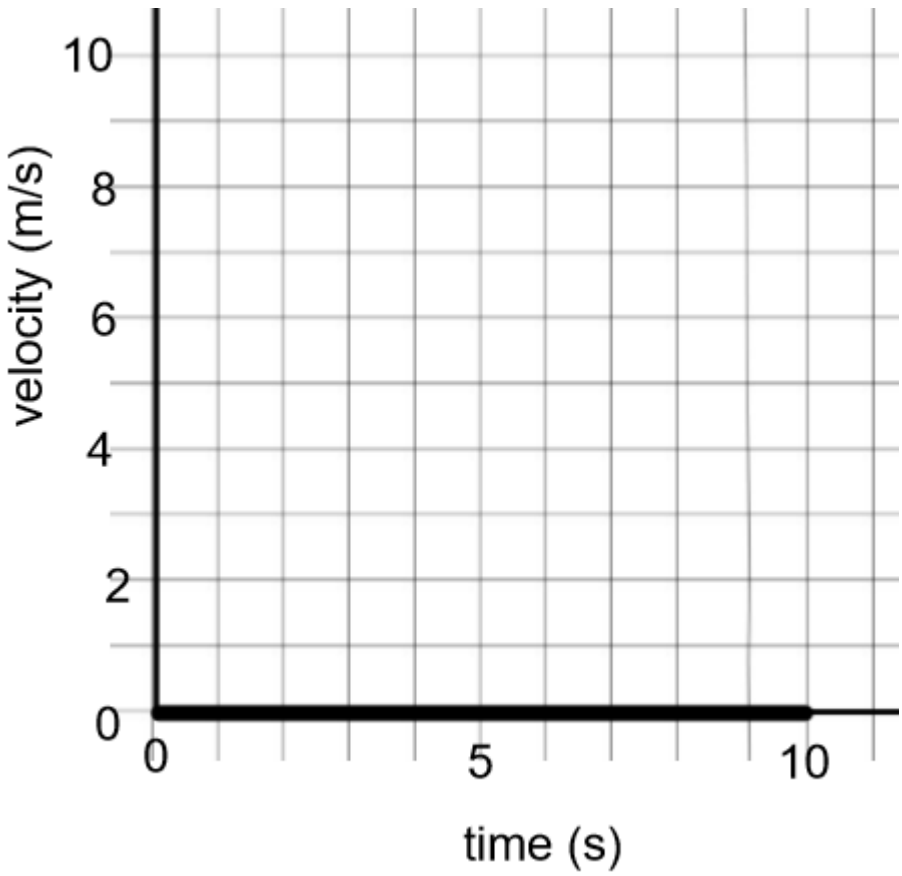
The motion of an object can be shown using a velocity-time graph:



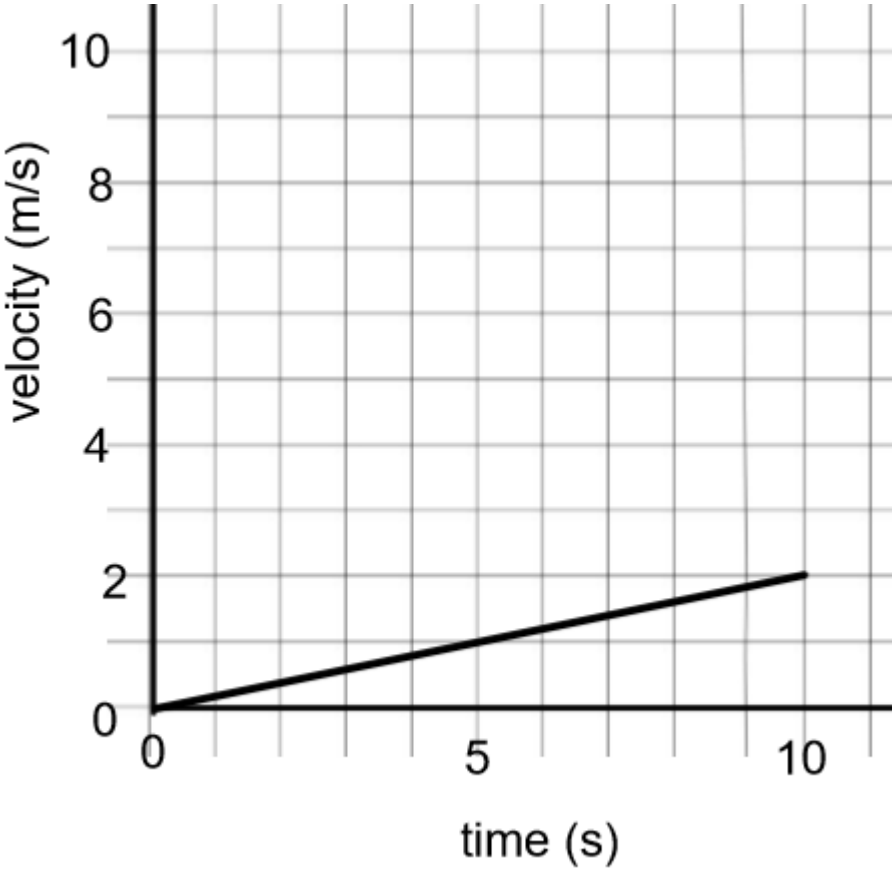
It's easy to confuse these with distance-time graphs, but it's important not to do so.



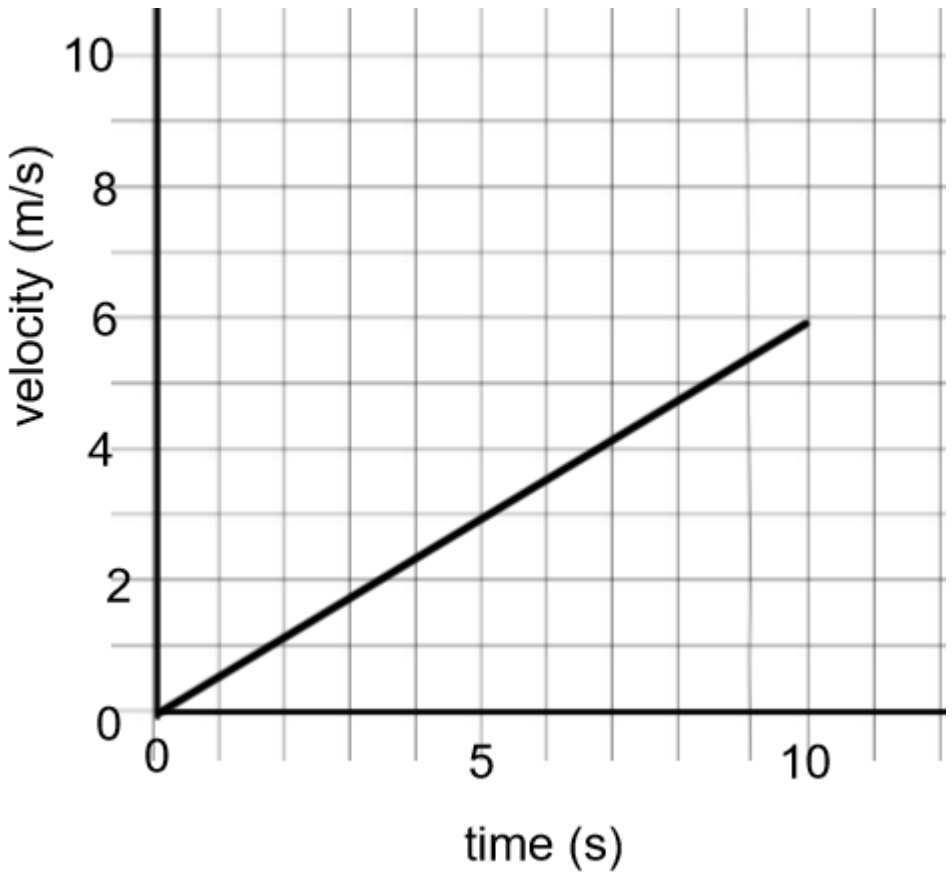
↑This graph shows an object moving at a constant velocity (steady speed). The gradient is 0 (horizontal line), so the acceleration is 0. Note this does **not** mean the object is stationary. In this example, the object is moving at a steady speed of 4m/s.



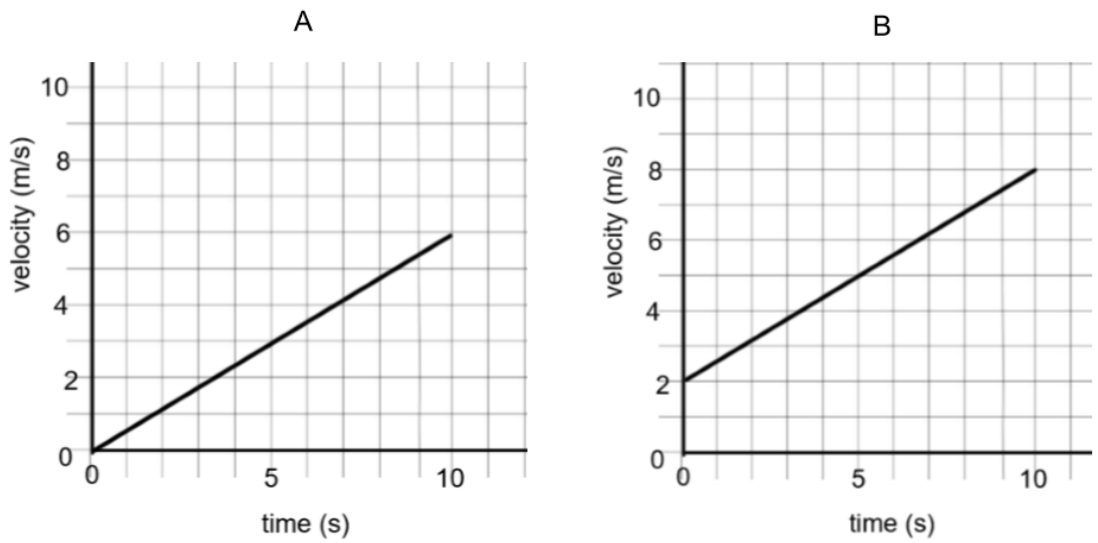
↑ This graph shows an object which is stationary. The line shows the velocity stays at 0 for the time shown. Note that stationary is a form of constant velocity: the gradient is 0 (the line is horizontal).



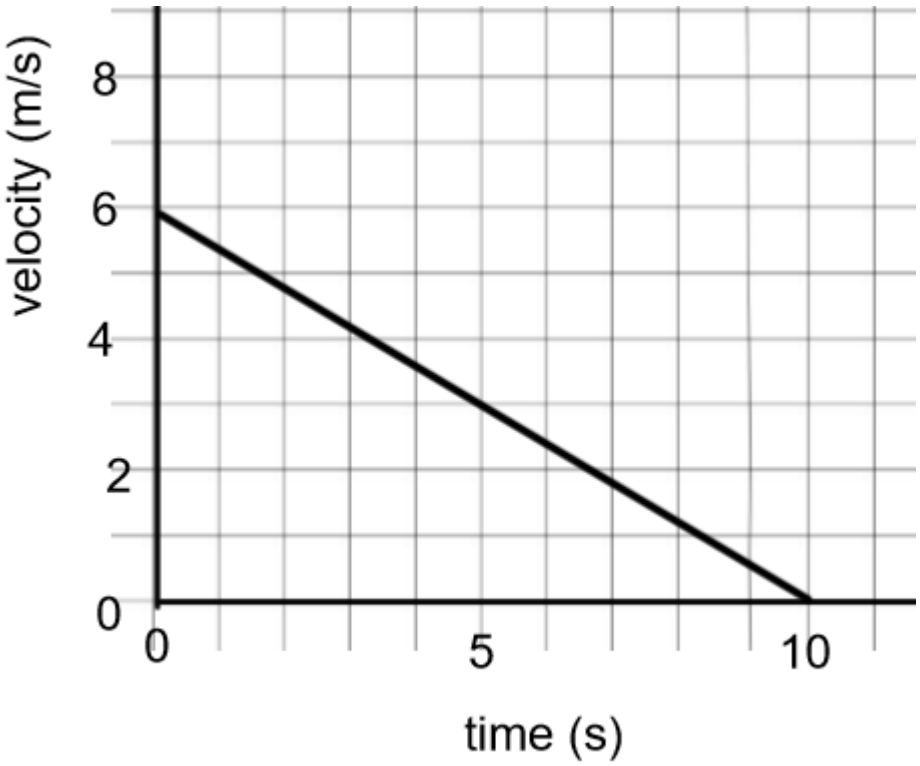
↑ This graph shows an object which is accelerating. The line has a gradient (it is sloping). The speed is changing. It starts off at 0m/s, and after 10 seconds it has increased to 2m/s.



↑ This graph shows an object with a higher acceleration, i.e. its speed is increasing more quickly. This object has only taken 3.5s to reach 2m/s. At 10s, its speed is 6m/s. The gradient is higher (the slope is steeper) because the acceleration is higher.



↑ The graphs above show two objects with the same accelerations but different starting (and therefore finishing) speeds.

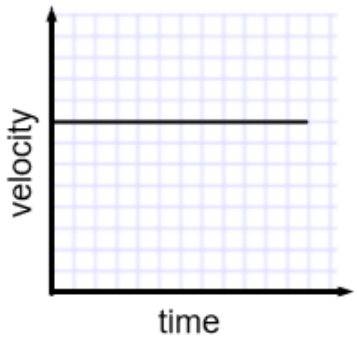


↑ This graph shows an object with negative acceleration (deceleration).

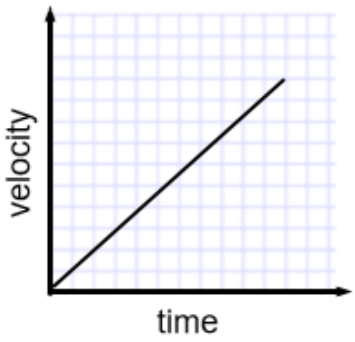
Questions:

State what each graph shows: acceleration, constant velocity, negative acceleration, or stationary:

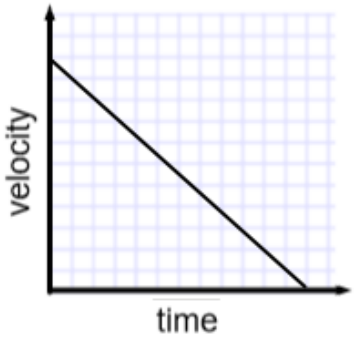
1.



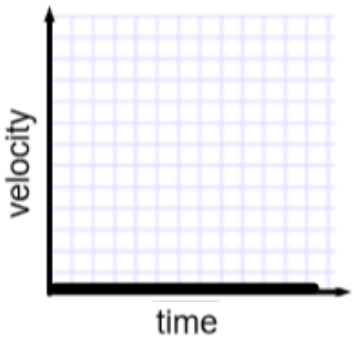
2.



3.



4.



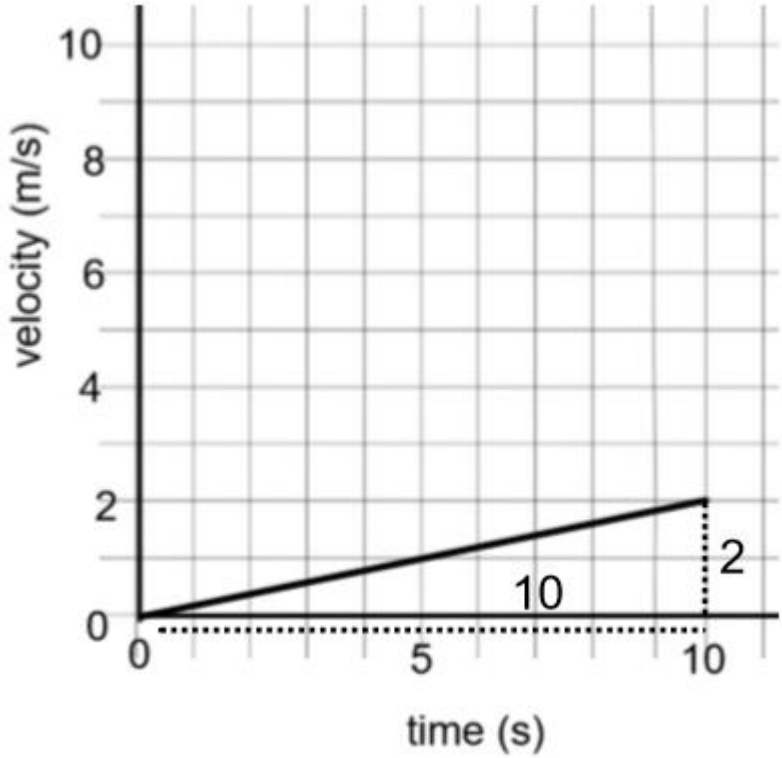
Finding acceleration from the gradient of a velocity-time graph

Because acceleration is change in velocity ÷ time, we can find acceleration of an object by finding the gradient of a velocity-time graph.

The first thing we need to know is that:

Change in velocity = final velocity – initial velocity

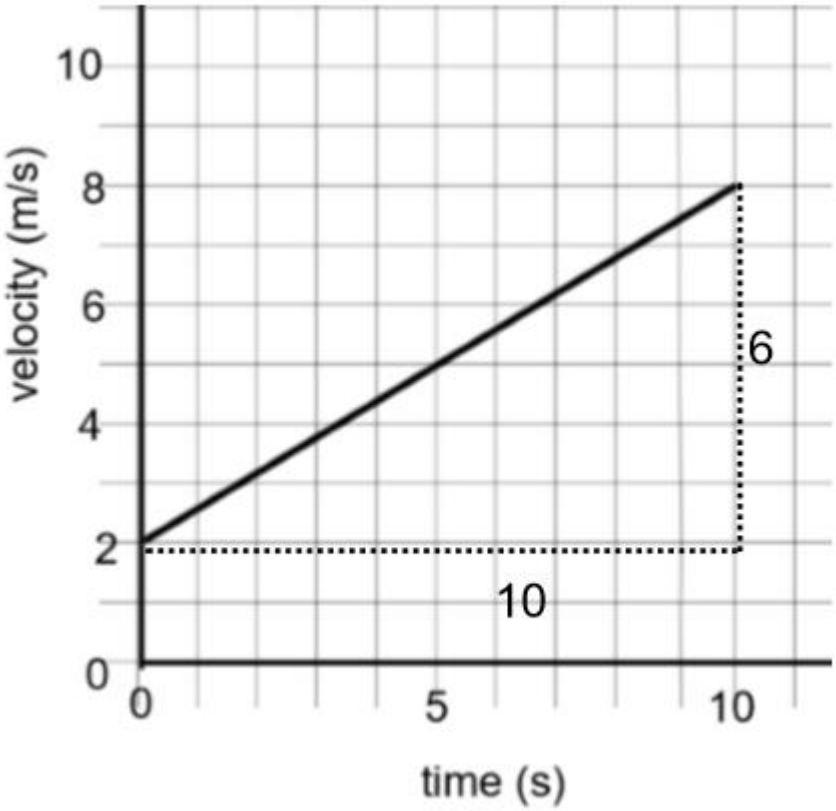
= end speed - starting speed



Change in velocity = 2 – 0 = 2m/s

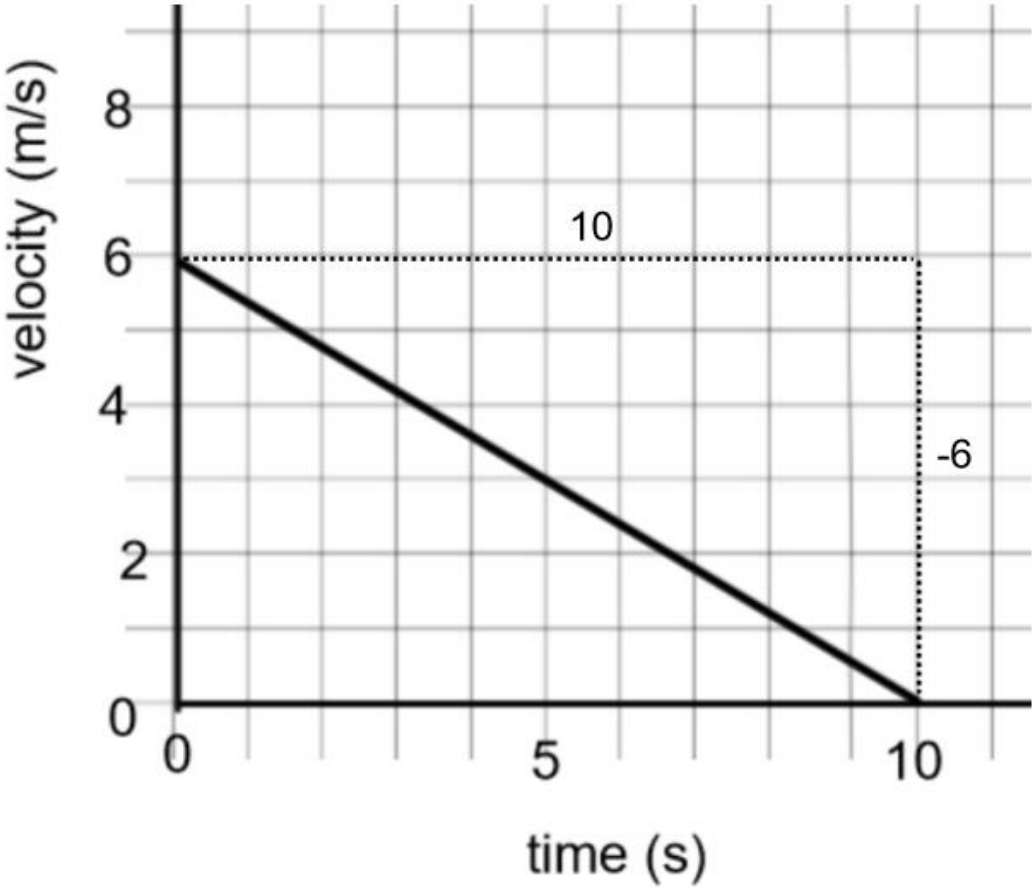
Acceleration = 2 ÷ 10 = 0.2m/s/s.

We have to be careful to use change in velocity, not final velocity:



Change in velocity = 8 – 2 = 6m/s
Acceleration = 6 ÷ 10 = 0.6m/s/s.

And if the gradient is negative:



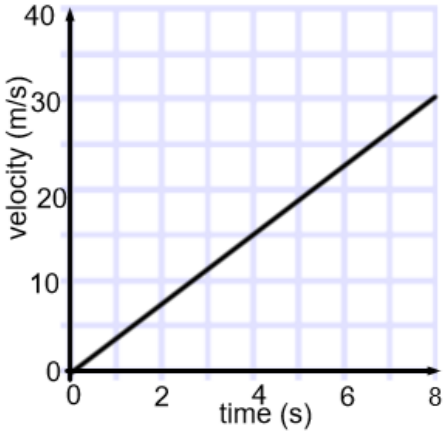
Change in velocity = 0 – 6 = -6

Gradient = -6 ÷ 10 = -0.6m/s/s
Acceleration = -0.6m/s/s

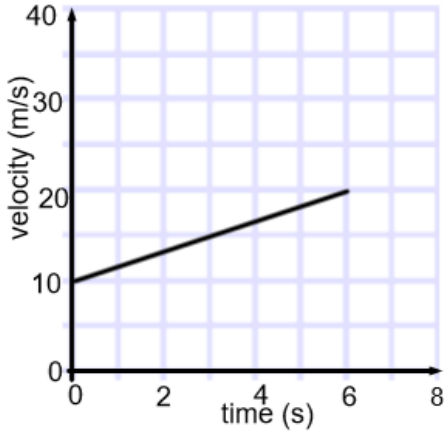
Questions:

Find the acceleration of the objects in the following graphs:

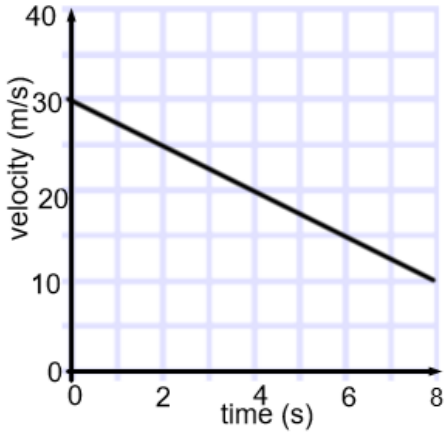
1.



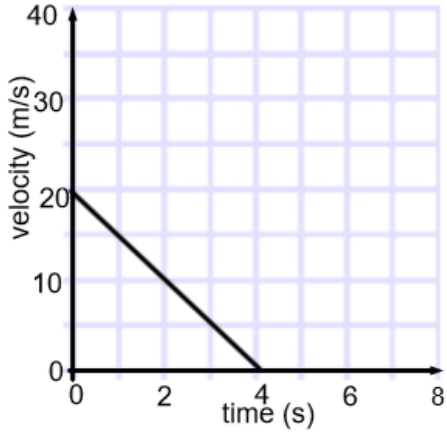
2.



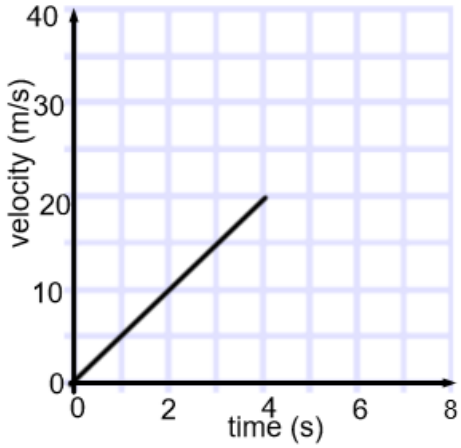
3.



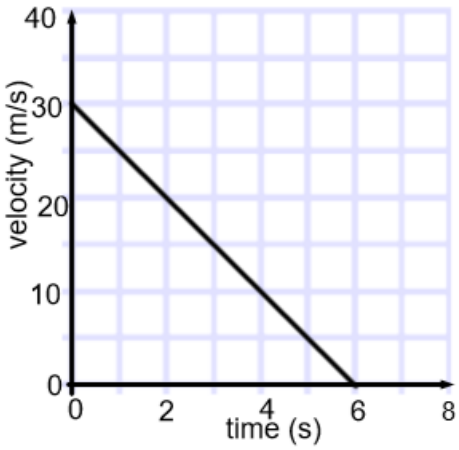
4.



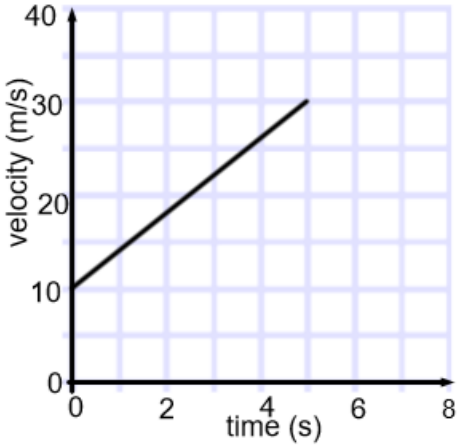
5.


40
30
20
10
0
velocity (m/s)
0
2
4
6
8
time (s)

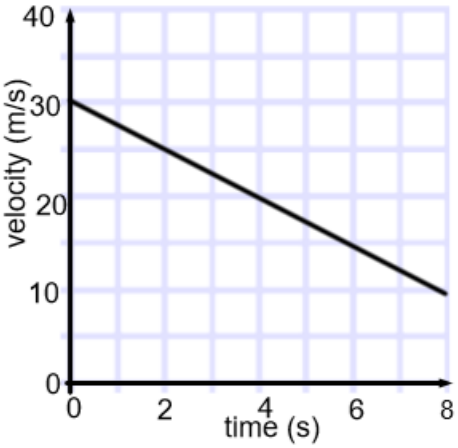
6.


40
30
20
10
0
velocity (m/s)
0
2
4
6
8
time (s)

7.

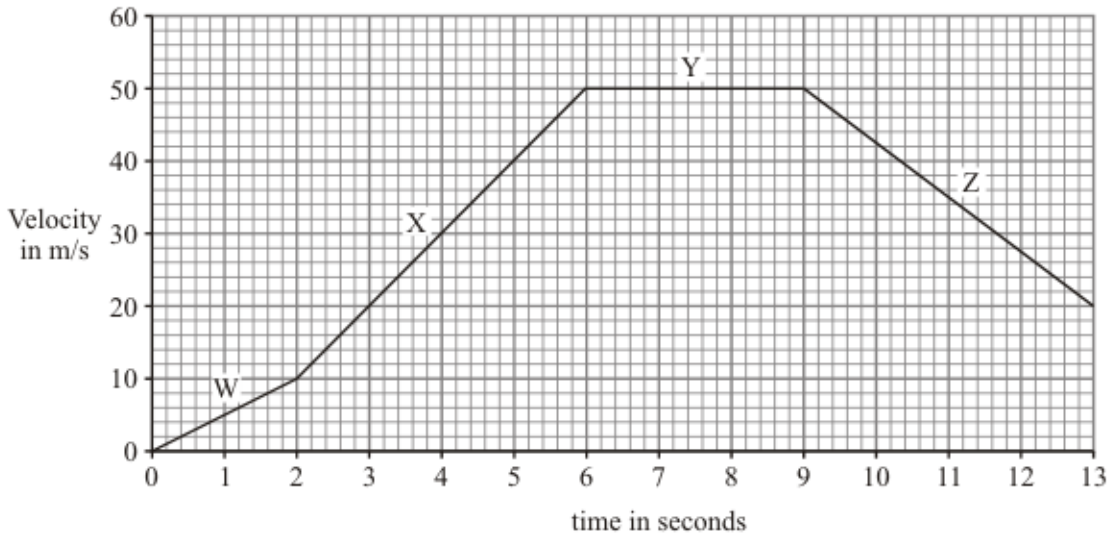

40
30
20
10
0
velocity (m/s)
0
2
4
6
8
time (s)

8.

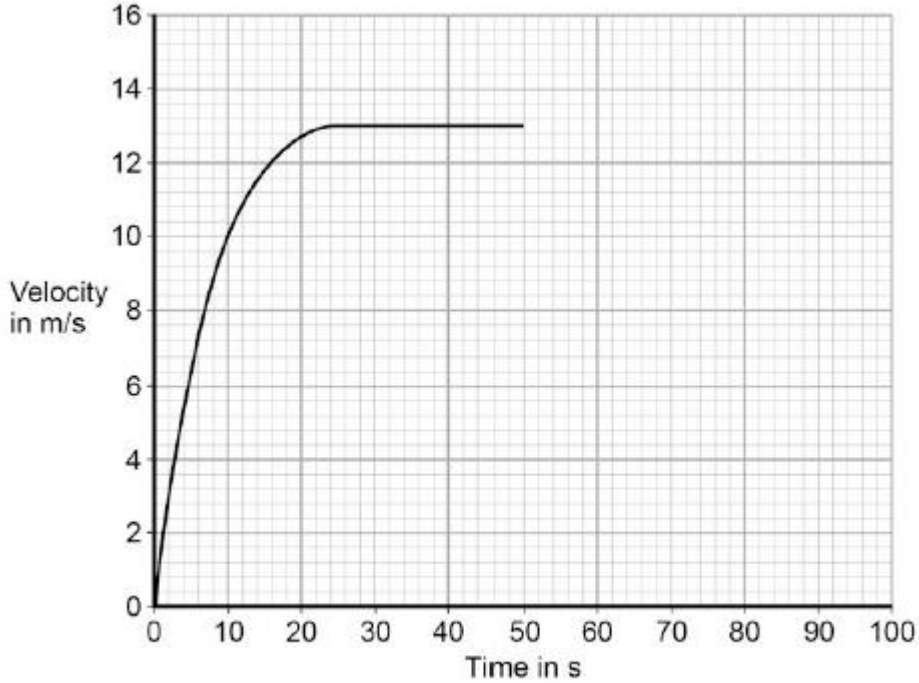

40
30
20
10
0
velocity (m/s)
0
2
4
6
8
time (s)

Velocity-time graphs practice

1. A car on a journey:

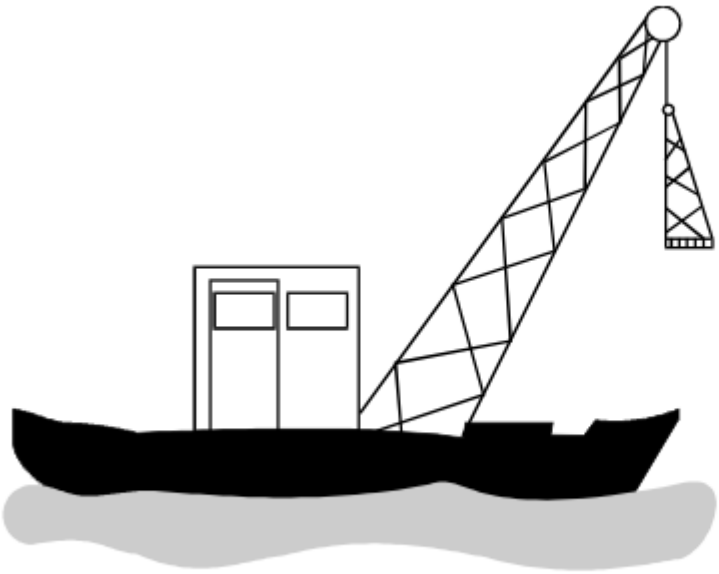


2. A skier sking down a slope:



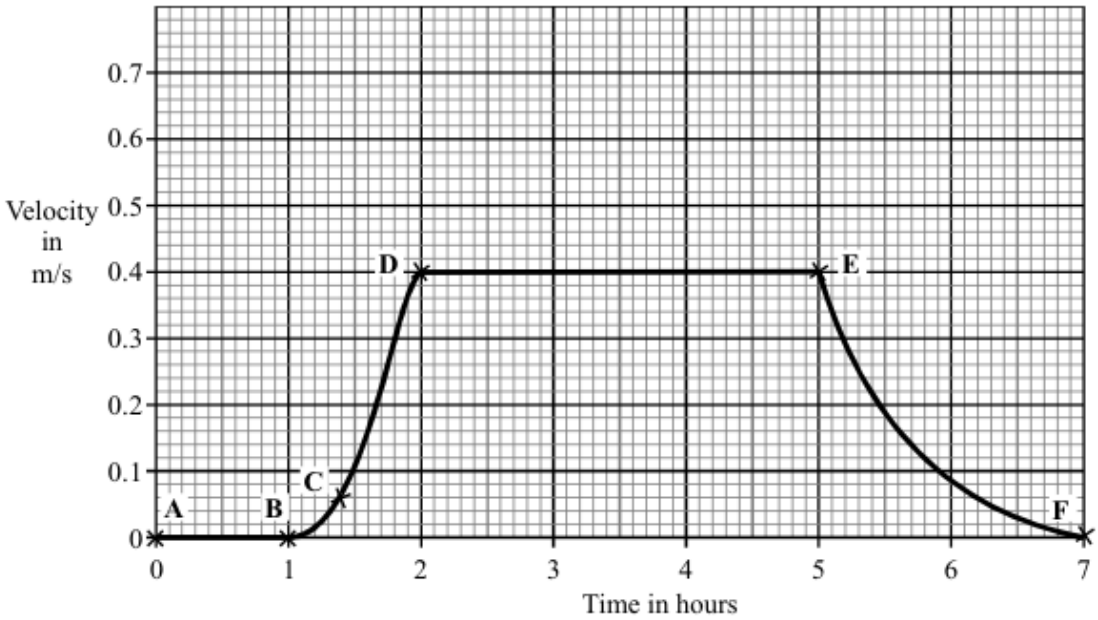
3.

A crane on a barge lifts a girder and then carries it along the river.

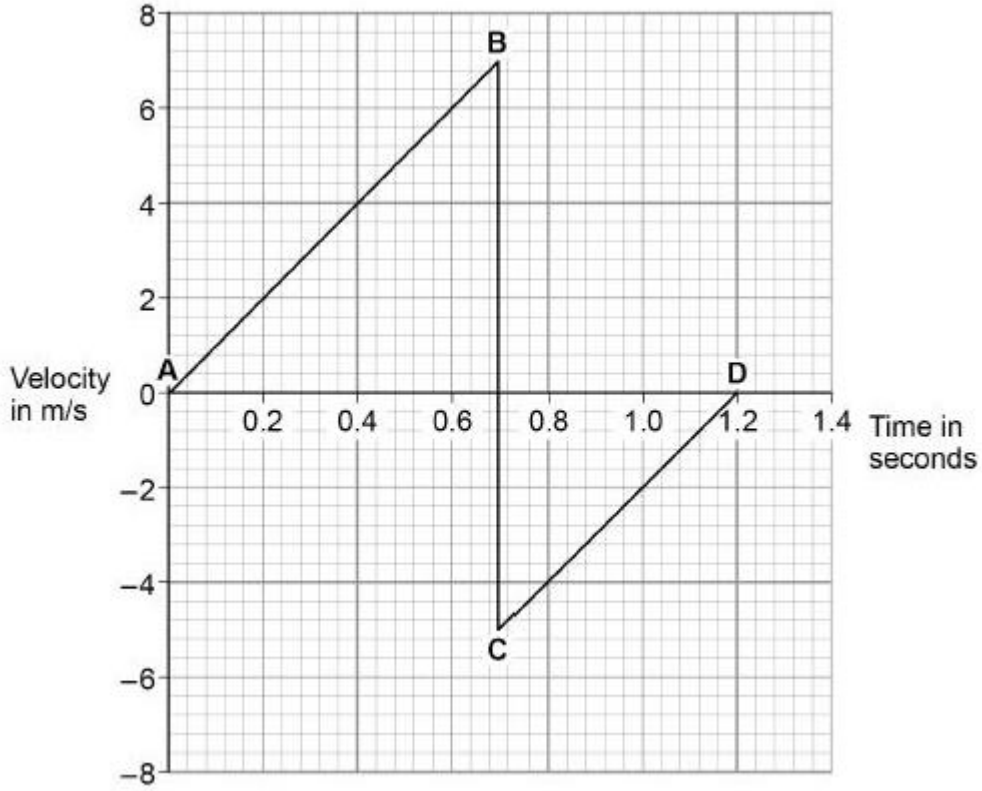


The girder has a weight of 1 000 000 N and is lifted to a height of 1500 cm.

Velocity of the barge after the girder has been lifted:



4. A ball being dropped and bouncing:



Calculating distance travelled for velocity-time graphs

The area under the graph is the distance travelled:

d/t x t = d

Find the distance travelled by the graphs on pages 29 – 36.

Acceleration formula 2

We can also use

$$v^2 = u^2 + 2as$$

Where:

v = final velocity
u = initial velocity
a = acceleration
s = displacement

This formula is given on the formula sheet.

This formula is useful when we don't know the time taken.

Questions:

1. Find the final velocity of an object whose initial velocity is 3m/s, is travelling with an acceleration of 4.8m/s^2, and travels for 24m.

2. Find the final velocity of an object whose initial velocity is 25m/s, is travelling with an acceleration of 0.5m/s^2, and travels for 200m.

3. Find the final velocity of an object whose initial velocity is 0.2m/s, is travelling with an acceleration of 1.9m/s^2, and travels for 140m.

4. An object begins with an initial velocity of 12m/s and ends with a velocity of 63m/s. If the distance travelled is 125m, what is the acceleration?

5. What acceleration is needed to bring an object from a speed of 1.8m/s to a speed of 15.9m/s over a distance of 35m?

6. An object is accelerated from 105m/s to 110m/s over a distance of 13m. What is the acceleration?

7. An object travels with an acceleration of $3.1m/s^2$. After it has travelled for 700m, it has reached a velocity of 207m/s. Find its initial velocity.

8. Find the initial velocity of an object travelling with an acceleration of $2.1m/s^2$, which after having travelled for 900m, has reached a velocity of 607m/s.

9. An object accelerates at a rate of $1.8m/s^2$. If it accelerates over a distance of 1600m, what must its initial velocity be in order to reach a final velocity of 560m/s?

10. An object has initial velocity 2m/s, final velocity 3m/s, and an acceleration of $0.25m/s^2$. Find the distance travelled.

11. What distance is travelled by an object travelling first at 6m/s, accelerating at $4m/s^2$ to a final velocity 30m/s?

12. An object has initial velocity 20m/s, final velocity 300m/s, and an acceleration of $18m/s^2$. Find the distance travelled.

13. An object has initial velocity 4m/s, final velocity 3m/s, and an acceleration of $-0.25m/s^2$. Find the distance travelled.

14. Find the final velocity of an object whose initial velocity is 400m/s, is travelling with an acceleration of $-14.8m/s^2$, and travels for 240m.

15. An object travels with an acceleration of $-3.1m/s^2$. After it has travelled for 700m, it has reached a velocity of 207m/s. Find its initial velocity.

16. What acceleration is needed to bring an object from a speed of 98m/s to a speed of 15.9m/s over a distance of 35m?

17. An object accelerates from rest to a velocity of 5m/s. If the acceleration is $0.9m/s^2$, calculate the distance travelled.

18. An object accelerates from rest to an unknown speed. If the object accelerates at $5m/s^2$ and travels for a distance of 15m, what is the final speed?

19. An object at rest begins to move and reaches a final speed of 165m/s after accelerating over a distance of 30m. What is the acceleration?

20. A moving object slows to a standstill at an acceleration of $-20m/s^2$. If the distance travelled is 60m, find the initial velocity of the object.
21. A car travelling at 35m/s crashes into a wall and is stopped over a distance of 10.2m. Find the acceleration of the car.

22. A stone is dropped and is travelling at a speed of 1.3m/s just before it hits the ground. If the acceleration is $-135m/s^2$, find the distance travelled into the ground by the stone.

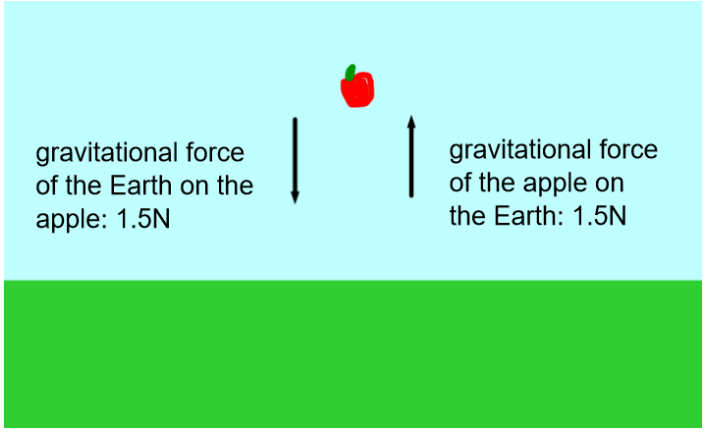
Newton's Third Law

Newton's Third Law states that:

"For every action there is an equal and opposite reaction"

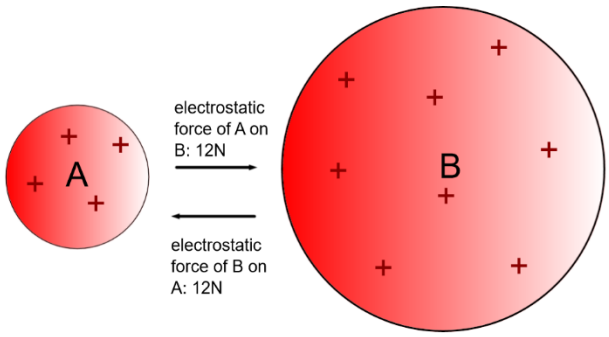
What this means is that because forces are interactions between pairs of objects, forces always come in pairs. We call these "Newtonian pairs".

Consider an apple falling from a tree:



The Earth pulls on the apple with a force of 1.5N. What's surprising is that the apple is also pulling on the Earth with a force of 1.5N. The apple is the thing we see accelerating, because its mass is small enough for 1.5N to create a significant acceleration. Newton's second law, F = ma is responsible for this.

Let's look at another example: two positively charged spheres, A and B:



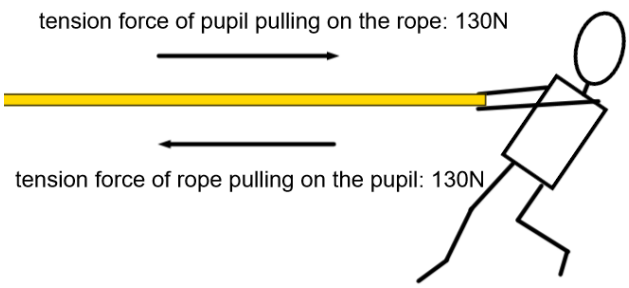
A repels B and B repels A with the same magnitude of force. This is surprising because B is larger in size and has more positive charge.

The two forces in an interaction are always:

- Equal size
- The same type of force (e.g. gravitational)
- Opposite direction
- Acting on the other object

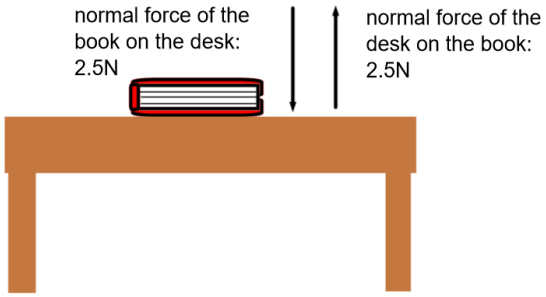
Some more examples:

When you pull on a rope, the rope pulls on you with a force of the same magnitude and opposite direction:



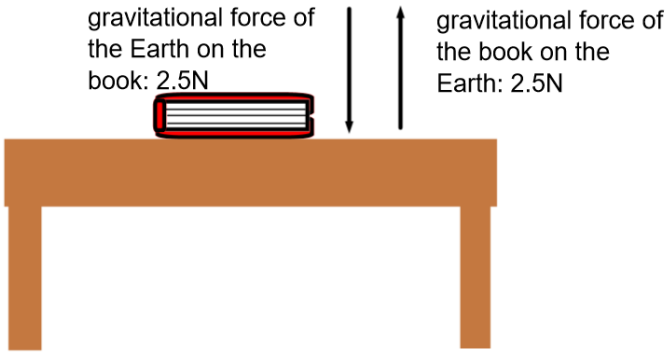
This is counter-intuitive: people often hold the misconception that because the rope isn't moving or alive, it can't be pulling – but this is wrong. If we imagine pulling on a spring, we can easily accept that the spring pulls back on us. The atoms in a rope and any other object act like springs – if we pull on them they pull back.

A book sitting on a desk pushes on the desk with a “normal” force. The table pushes back on the book with a “normal” force of the same magnitude and opposite direction:



Again, people often hold the misconception that the table cannot be pushing on the book, as it is not alive or moving – but this is wrong. (Note that people have less trouble believing that the book pushes on the table – perhaps because if we place our hand under the book we can feel it pushing.) Again, if we imagine pushing down on a spring, we can imagine the spring pushing back up on our hand – and the atoms in the table behave like springs, pushing back on anything that pushes on them.

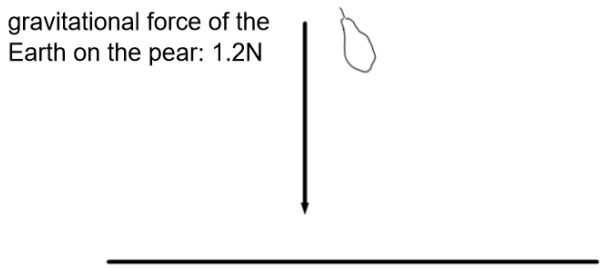
Note we have not considered the weight of the book here, if we were to consider the weight of the book it would be:



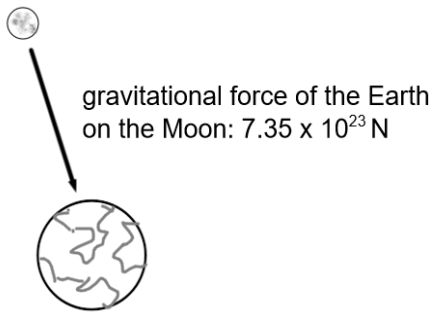
Questions

1. State Newton's third law.
2. What is a "Newtonian pair"?
3. Copy the diagram showing the gravitational forces between the Earth and an apple.
4. Why doesn't the Earth appear to be affected by the force exerted on it by the apple?
5. Copy the diagram showing the tension forces on the rope and the person pulling the rope.
6. Copy the diagram showing the normal forces between the book and the table.
7. What are the four characteristics of the forces in Newtonian pairs?
8. Copy and complete the following diagrams to show the second force in the Newtonian pair:

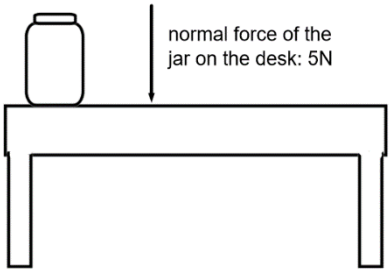
a.

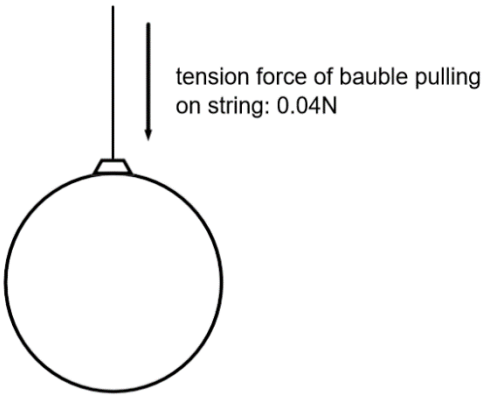


b.



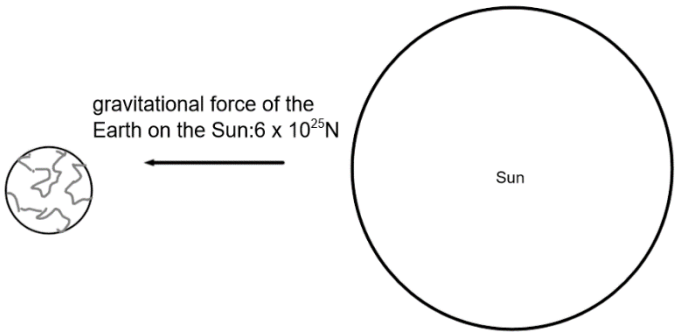
c.


normal force of the
jar on the desk: 5N

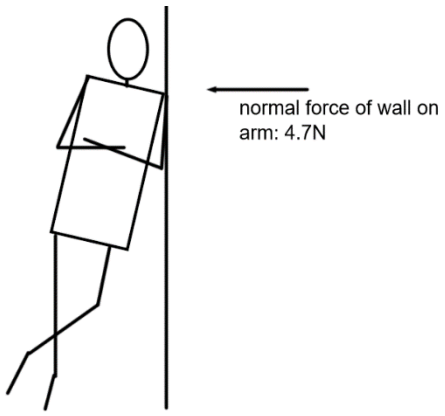

tension force of bauble pulling
on string: 0.04N

d.

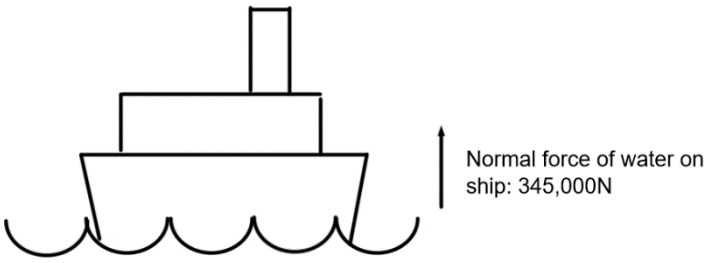
e.


gravitational force of the
Earth on the Sun:6 x 10^25N
Sun

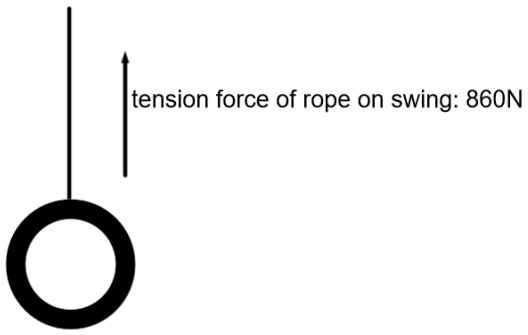
f.


normal force of wall on
arm: 4.7N

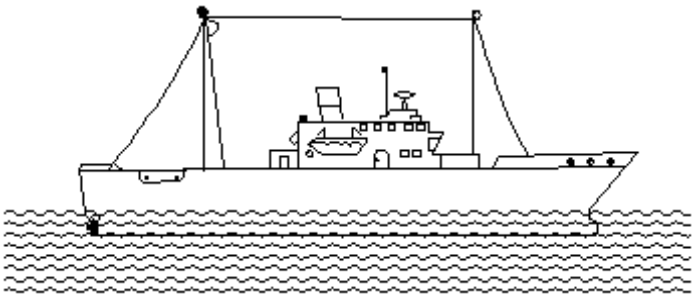
g.


Normal force of water on
ship: 345,000N

h.

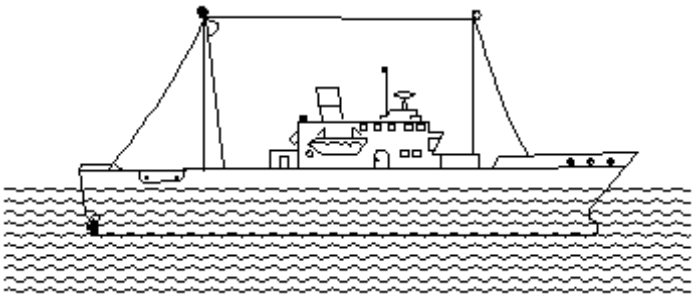

tension force of rope on swing: 860N

2. The diagram below shows an empty cargo ship. It is not moving.



(a) The water exerts a force on the ship. In which direction does this force act?

(b) The diagram below shows the same cargo ship. This time it has a full load of cargo.



(i) How does the force exerted by the water on the ship change as the ship is loaded?

(ii) Why has the force exerted by the water changed?

Newton's Second Law

A resultant force on an object free to move will cause the object to accelerate.

Acceleration is a change in velocity so it can be an increase or decrease in speed, or a change in direction.

The larger the mass of an object, the more difficult it is to get it to change its velocity: a larger force is needed to accelerate it. This relationship is shown in the formula:

Force = mass x acceleration

Acceleration = force ÷ mass

Questions

1. Mass = 12kg

Acceleration = $25m/s^2$

Force = ?

2. Mass = 100g

Acceleration = $6.3m/s^2$

Force = ?

3. Mass = 308kg

Acceleration = $0.56km/s^2$

Force = ?

4. Acceleration = $7.9m/s^2$

Force = 91N

Mass = ?

5. Acceleration = 2.4m/s^2

Force = 75N

Mass = ?

6. Acceleration = 0.6m/s^2

Force = 5,000N

Mass = ?

7. Force = 13kN

Mass = 58kg

Acceleration = ?

8. Force = 960N

Mass = 21kg

Acceleration = ?

9. Force = 0.4N

Mass = 76kg

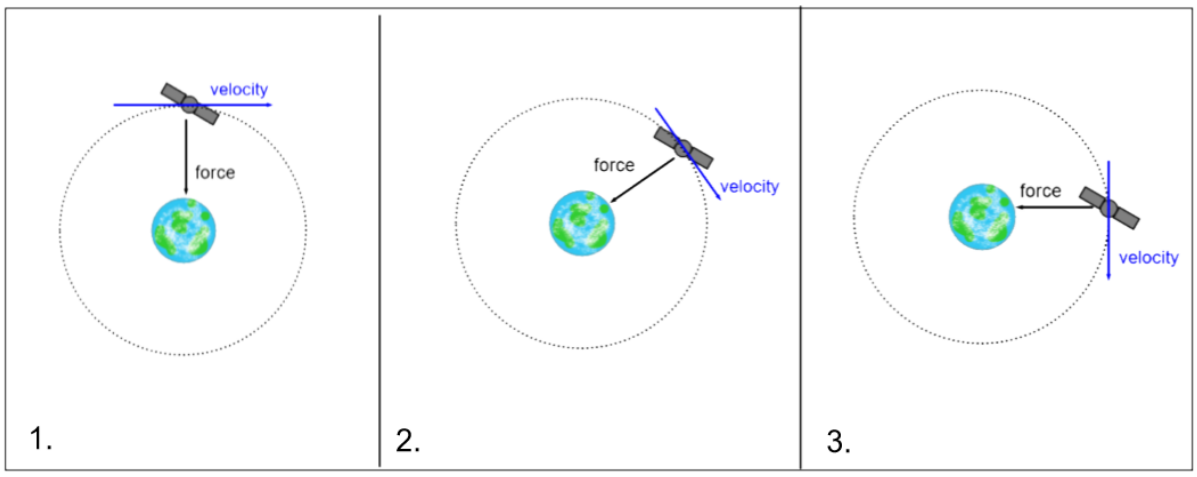
Acceleration = ?

10. What force is needed to accelerate a mass of 10kg at a rate of 2m/s^2?
11. An unknown force causes a mass of 180kg to accelerate at a rate of 90cm/s^2. Calculate the magnitude of the force.
12. Nasa wish to accelerate a satellite of mass 460kg at a rate of 98m/s^2. What force will they need?
13. An unknown mass is accelerated at 43m/s^2 when a force of 700N is applied to it. Find the mass.

14. What is the mass of an object if a force of 57N is needed to accelerate it by $0.6m/s^2$?
15. A force of 0.5N causes an object to accelerate at $24cm/s^2$. Find the mass of the object.
16. A force of 430N is applied to a mass of 2.8kg. Find the acceleration.
17. What acceleration will an object of 5kg undergo if a force of 90N is applied?
18. Find the acceleration of a 360kg object when a 0.2N force is applied.

Circular motion (HT)

Uniform circular motion involves an object with a tangential motion and a force acting towards the centre of the circle. Consider a satellite orbiting the Earth:



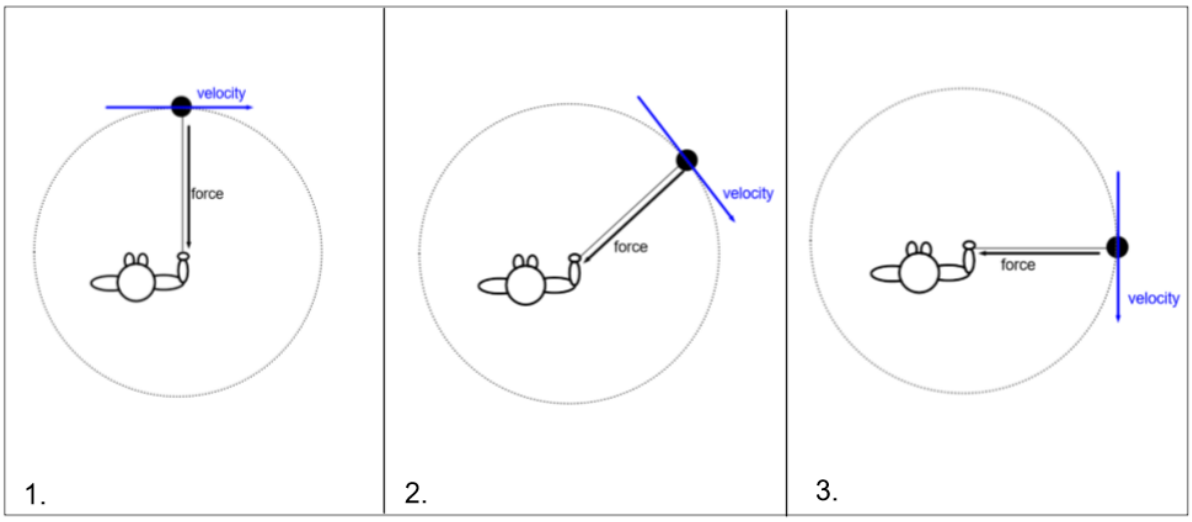
As the satellite travels on its circular path around the Earth, its velocity is constantly changing as it is changing direction. At any given moment, the satellite's velocity is a tangent to the circle.

There is a force constantly acting towards the centre of the circle: the force of the Earth's gravity on the mass of the satellite.

This force on the satellite causes it to accelerate: its direction constantly changes although its speed is constant.

The acceleration is towards the centre of the circle: the satellite's direction is constantly changing towards the centre of the circle.

In physics we do a demonstration with a bung on a string which we whirl around in a circle:



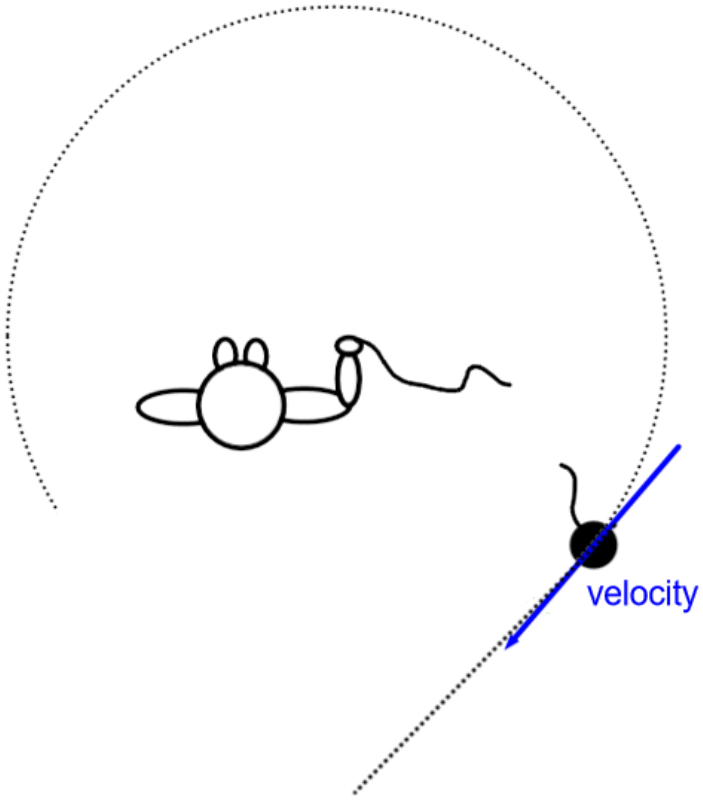
As the bung travels on its circular path around the pupil's hand, its velocity is constantly changing as it is changing direction. At any given moment, the bung's velocity is a tangent to the circle.

There is a force constantly acting towards the centre of the circle: the force of the string's tension on the bung.

This force on the bung causes it to accelerate: its direction constantly changes although its speed is constant.

The acceleration is towards the centre of the circle: the bung's direction is constantly changing towards the centre of the circle.

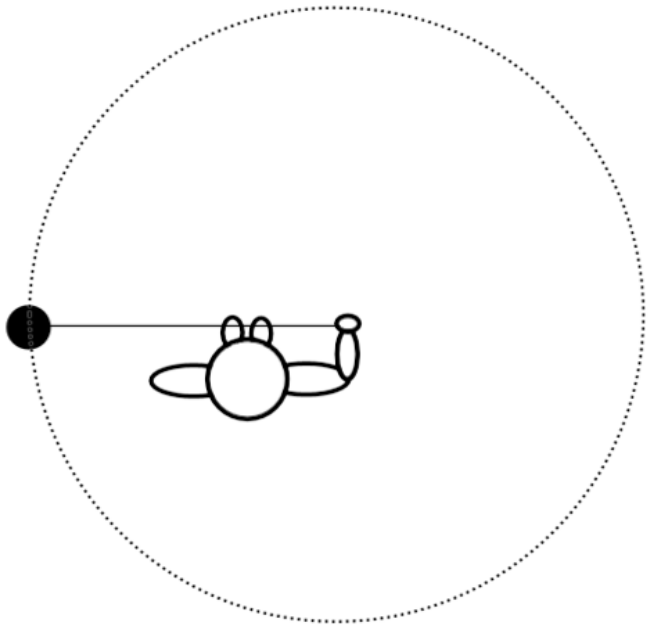
If the string breaks, the bung will fly off in a straight line – it will **not** continue in its circular path as the force is no longer there to accelerate it (change its direction).



Questions

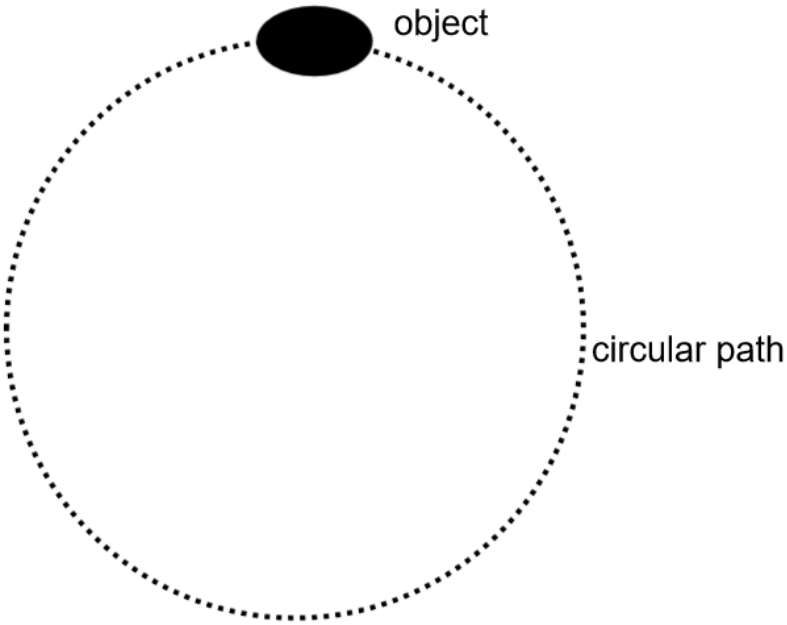
1. Copy the diagram showing circular motion of a satellite orbiting the Earth.
2. Draw stages 4, 5 and 6 for the satellite.
3. What force causes the satellite to travel with circular motion?
4. In what direction does this force act?
5. Why is circular motion at a constant speed classed as acceleration?
6. What direction is the acceleration of the satellite at any given moment?
7. What direction is the velocity of the satellite at any given moment?
8. What force causes the whirled bung on a string to travel with circular motion?

9. This diagram shows the bung at a later stage (clockwise motion). Copy the diagram and draw and label arrows to show the force on the bung, its acceleration and its velocity.



10. State what will happen if the string breaks in the whirling bung experiment and explain why this will happen.
11. A remote control car is tied to a string and the other end of a string is nailed to the floor. A pupil holds down “forwards” on the controller. What will happen and why?

12. Christine observes an object moving with circular motion in an anti-clockwise direction. She knows that at any given moment this object has a force acting on it, a velocity and an acceleration. Copy the diagram and state the direction of each of these quantities. You may add arrows to the diagram if you wish.



Newton's First Law

Newton's first law states that:
"When the resultant force on an object is zero:
A stationary object will remain stationary;
A moving object will continue to move at a constant velocity."

This means that a resultant force is needed to:

- start a stationary object moving
- slow down a moving object
- speed up a moving object
- stop a moving object
- change the direction of a moving object

This is counter-intuitive for a couple of reasons.

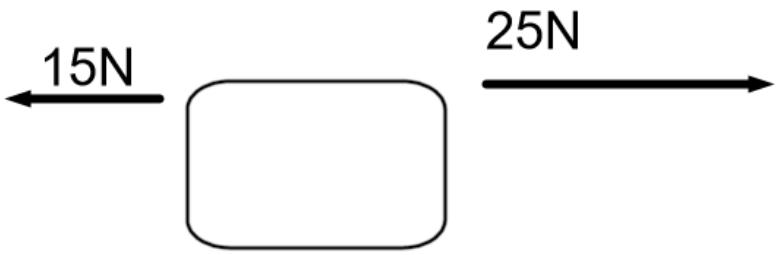
1. It is natural to think that if an object is moving it must have a resultant force acting on it – but this is false. Often in everyday life we need to apply a force to an object to keep it moving, such as sliding a book along a desk, but this is because we need to overcome the force of friction.

2. When objects keep moving, such as a kicked ball, it is natural to think they have a force propelling them forward – but this is false.

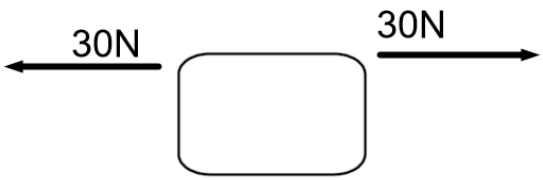
If we look at a free-body diagram of an object, showing the forces on it, it is impossible to tell whether the object is moving or stationary. What we can tell is whether it has a constant velocity or is accelerating. In everyday life we tend to distinguish between "moving" and "not-moving". In physics, we distinguish between "constant velocity" and "accelerating".

Questions

1. State Newton's First Law.
2. This object is accelerating. Why?



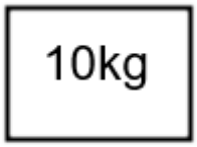
3. This object is either at rest or travelling at a constant velocity. Why?



Inertia (HT)

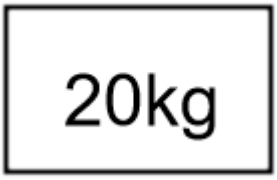
Objects with mass have a tendency to resist changes to their motion. This tendency is called inertia.

Consider a 10kg block at rest:



It will be fairly difficult to get it moving at 5m/s.

Once it is moving at 5m/s, it will be fairly difficult to get it to slow down or stop.



A 20kg block will be twice as difficult to get moving at the same speed. it will also be twice as difficult to stop from the same speed.

The 20kg block has twice as much inertia as the 10kg block.

Closely related is the concept of **inertial mass**. Inertial mass is a measure of how difficult it is to change the velocity of an object.

inertial mass = force ÷acceleration

Questions

1. Define "inertia"
2. Which has more inertia, a moving 2kg block or a stationary 6kg block?
3. What does "inertial mass" measure?
4. Give the formula for inertial mass.
5. An object requires 200N to accelerate it by 5m/s. Calculate its inertial mass in kg.

Terminal Velocity

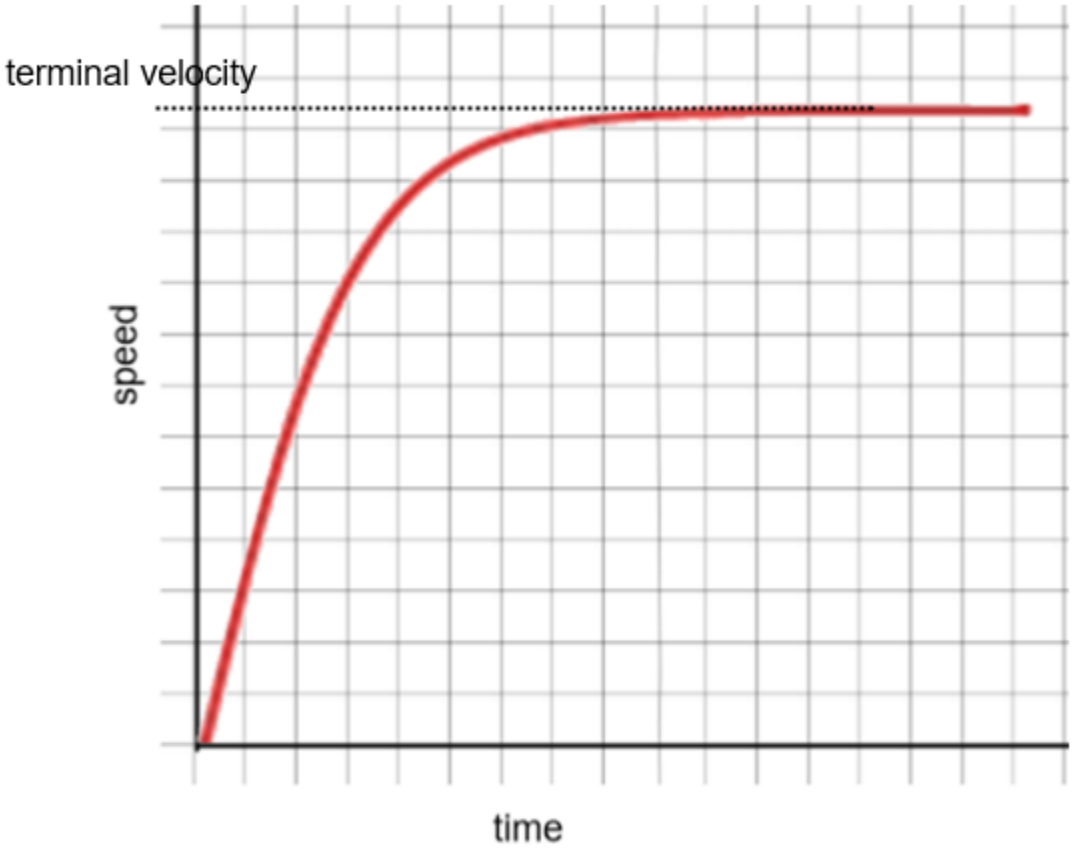
Consider an object falling through a fluid such as air.

At first, the object will accelerate due to the gravitational force (its weight).

As the object's speed increases, the air resistance (reaction force) increases.

After a time, the reaction force is equal to the weight. At this point the resultant force on the object becomes 0.

The object is now in equilibrium – it no longer accelerates but continues at a constant velocity. This is called the terminal **velocity**.

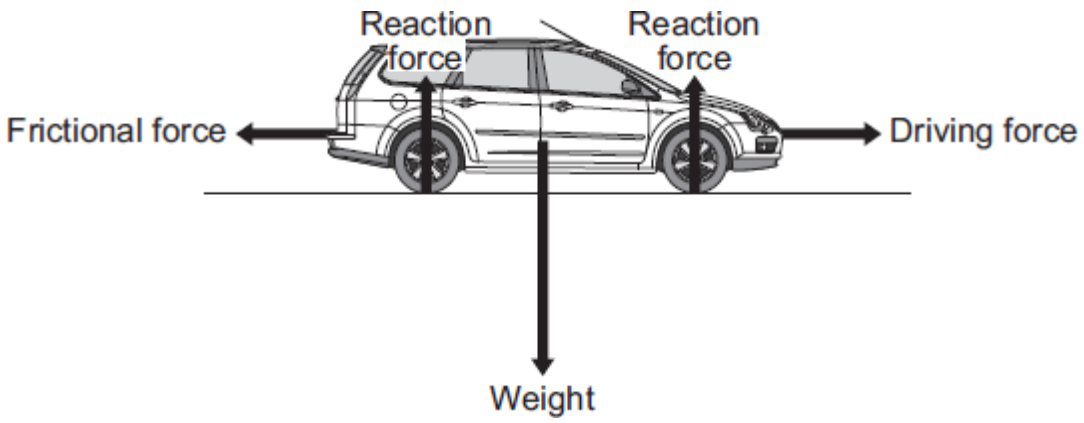


Questions

1. An object falling through a fluid such as air will accelerate at first and then reach a constant velocity. What name is given to this velocity?
2. What force causes an object to accelerate when falling?
3. What force causes an object falling through air to stop accelerating?

4. Why does air resistance increase while the object is falling?
5. Describe the forces on the object when the object is accelerating.
6. Describe the forces on the object when it has reached terminal velocity.
7. Copy and label the graph of speed over time for an object reaching terminal velocity.
8.

The diagram shows the forces acting on a car. The car is being driven along a straight, level road at a constant speed of 12 m/s.



(a) The driver then accelerates the car to 23 m/s in 4 seconds.

Calculate the acceleration of the car.

(b) Describe how the horizontal forces acting on the car change during the first **two** seconds of the acceleration.

9.

The manufacturer of a family car gave the following information.

Mass of car 950 kg

The car will accelerate from 0 to 33 m/s in 11 seconds.

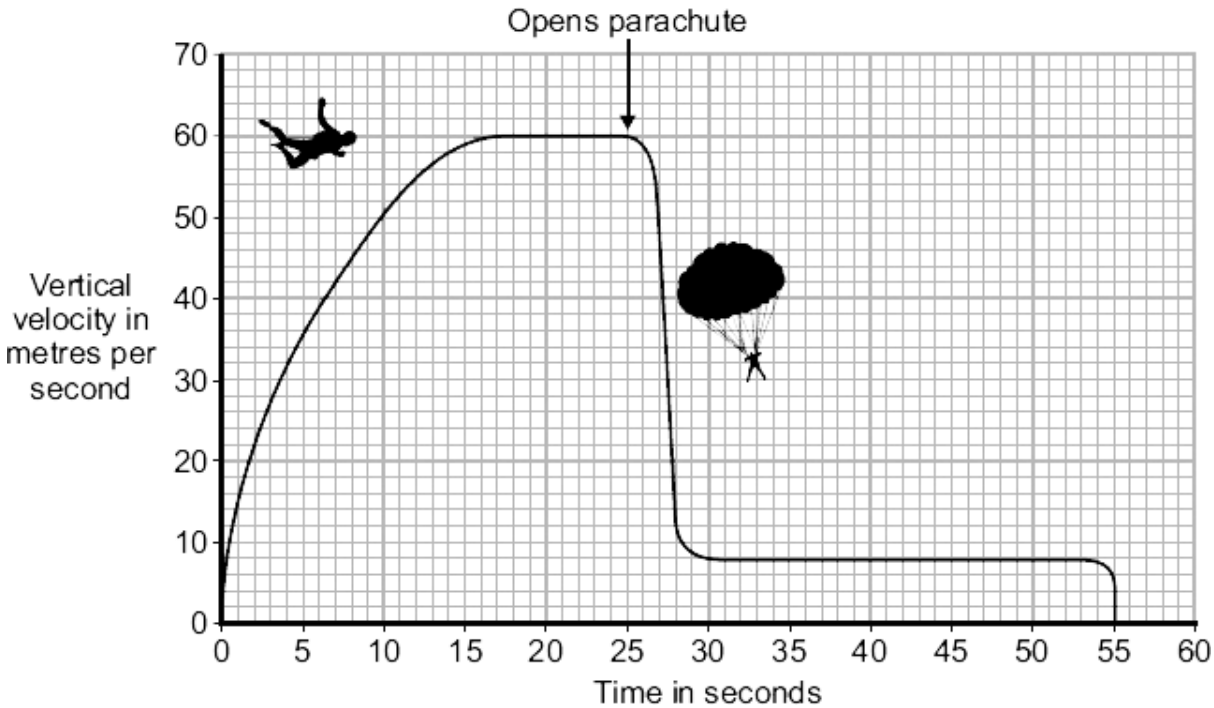
(a) Calculate the acceleration of the car during the 11 seconds.

(b) Calculate the force needed to produce this acceleration.

(2)

(c) The manufacturer of the car claims a top speed of 110 miles per hour. Explain why there must be a top speed for any car.

10. The graph shows how the vertical velocity of a parachutist changes from the moment the parachutist jumps from the aircraft until landing on the ground.



Using the idea of forces, explain why the parachutist reaches a terminal velocity and why opening the parachute reduces the terminal velocity.

Stopping distance

When a driver sees something that means they need to break, nerve signals must travel around the body in order for the foot to press the brake pedal. The foot does not press the brake immediately but a short time after the hazard is seen. This time is called the thinking time.

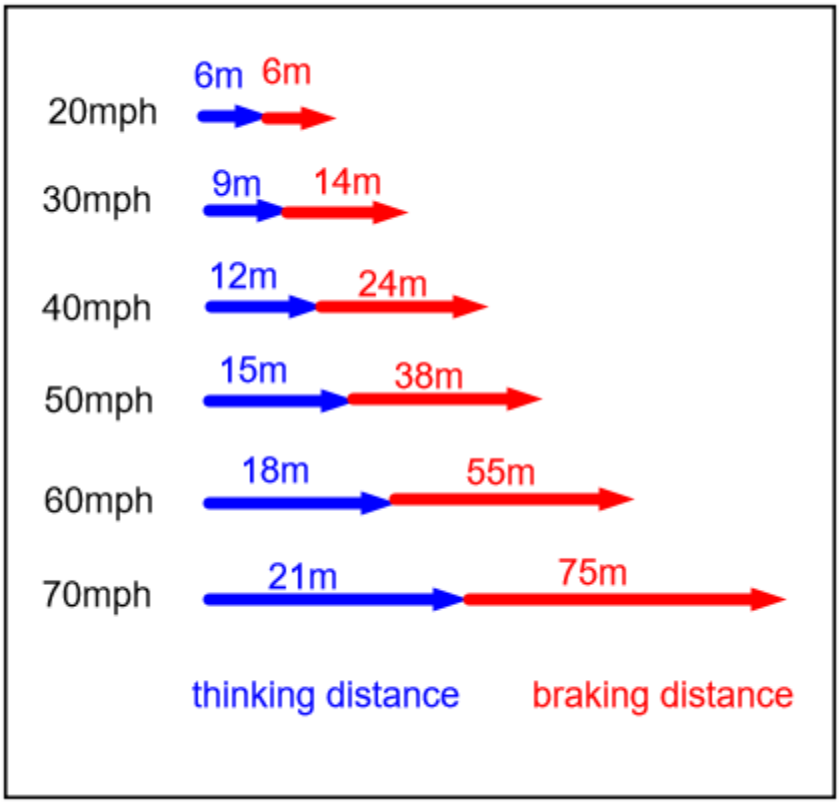
When a car brakes, the breaking force acts on it to reduce its velocity to zero. The car does not stop immediately but takes an amount of time to come to a stop. This time is called the braking time.

The car travels during these times. The distance travelled during "thinking" is called the thinking distance. Thus we can define thinking distance as "the distance travelled while the need to brake is processed by the nervous system".

The distance travelled during braking is called braking distance. Thus we can define braking distance as "The distance travelled while the braking force decelerates the car to 0 velocity".

The total of these two distances is called the stopping distance.

Thinking distance + braking distance = stopping distance



For a given braking force, the greater the speed of the vehicle, the greater the stopping distance.

Questions

1. What is “thinking distance”?
2. What is “braking distance”?
3. What is “stopping distance”?
4. For a given braking force, what is the relationship between speed and stopping distance?

Reaction time

When a driver sees and reacts to a hazard, the following chain of events takes place:

Light signal from hazard enters the eye → nerve signal to brain → processing in brain → signal to muscles in foot → muscles contract in foot

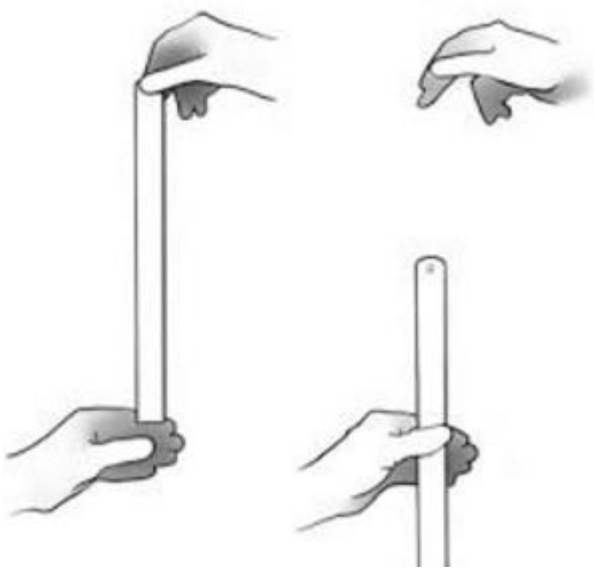
The time taken by this chain of events is called the reaction time. It can also be called the thinking time.

Reaction times vary from person to person. Typical values range from 0.2 s to 0.9 s.

A driver's reaction time can be affected by tiredness, drugs and alcohol, or distractions.

Measuring reaction times

One way of measuring a person's reaction time is to have them attempt to grip a falling ruler. The shorter the distance fallen by the ruler, the quicker the reaction time.



Questions

1. Write out the chain of events that takes place during "reaction time".
2. Give 3 typical values for reaction time.
3. State four things that can affect a driver's reaction time.
4. Describe the "falling ruler" method for measuring reaction times

5. Some pupils carried out an experiment on factors affecting reaction time. Write a conclusion from their results:

Conditions	Reaction time (s)		
	Test 1	Test 2	Test 3
Control	1.2	1.4	1.3
Distracted – talking	1.8	1.9	1.9
Tired	1.7	1.7	1.9
Caffeine	1.0	0.9	0.8

Momentum (HT)

Consider an object at rest.

Any object that moves has the property of momentum. Momentum can be found using the formula

momentum = mass x velocity

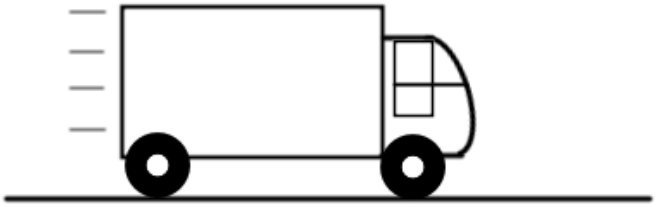
$$\rho = mv$$

momentum in kgm/s (kilogram metres per second)

mass in kg

velocity in m/s.

Momentum can be described as the amount of movement an object has. If an object is moving faster it has more movement, and a more massive object has more movement than a less massive one at the same speed. Because momentum depends on both mass and velocity, it is possible for a lorry and a bullet to have equal momentum:



mass = 3,000kg

velocity = 0.02m/s

momentum = 60 kgm/s

mass = 0.04kg

velocity = 1,500m/s

momentum = 60kgm/s

Questions

1. Write the formula for momentum
2. What are the units of momentum?
3. What two things is momentum a product of?
4. mass = 4kg

velocity = 12m/s
momentum = ?

5. mass = 118kg

velocity = 0.2m/s
momentum = ?

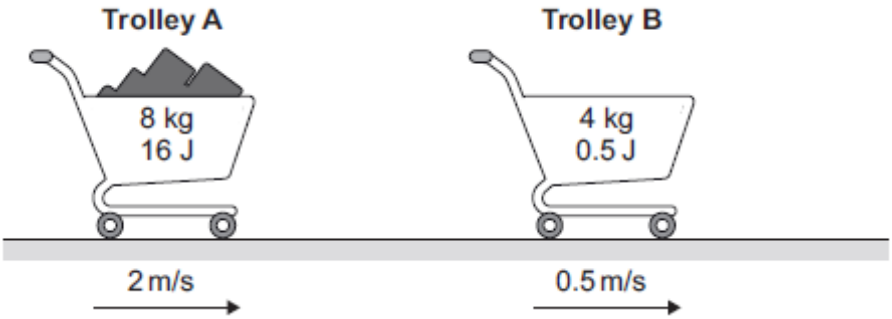
6. mass = 45kg
velocity = ?
momentum = 90kgm/s

7. mass = 200kg
velocity = ?
momentum = 67kgm/s

8. mass = ?
velocity = 16m/s
momentum = 64kgm/s

9. mass = ?
velocity = 58m/s
momentum = 232kgm/s

10. Find the momentum of an object of mass 23kg travelling at a velocity of 4m/s
11. A trolley of mass 1.2kg has a velocity of 5.1m/s. Find its momentum.
12. An object has a momentum of 58kgm/s and a mass of 12kg. Find its velocity.
13. Find the velocity of a cart with a momentum of 90kgm/s if its mass is 63kg.
14. What is the mass of a truck if it has a momentum of 18,000kgm/s at a velocity of 3.4m/s?
15. Find the mass of an object with mass 37kg and momentum 780kgm/s.
16. Find the momentum of each trolley:



Conservation of momentum

The principle of conservation of momentum:

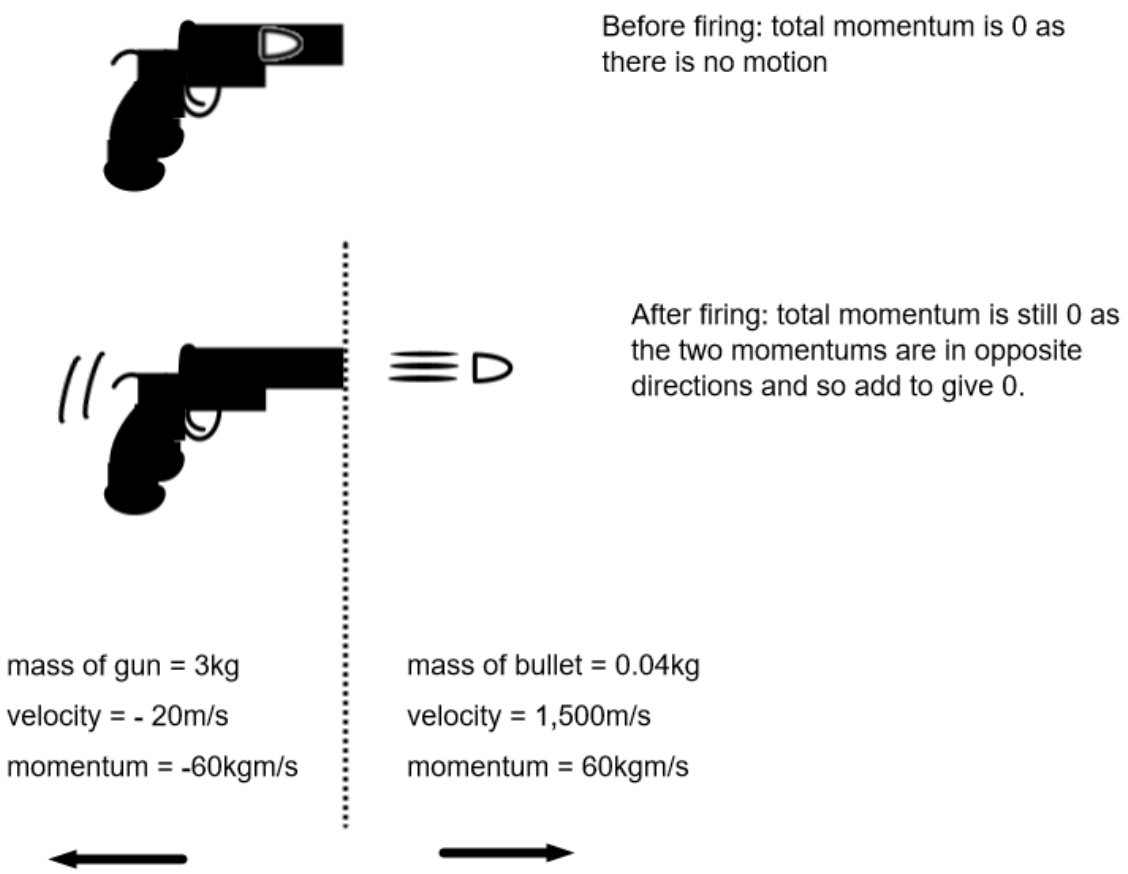
"In a closed system, the total momentum before an event is equal to the total momentum after the event."

A closed system is an object or group of objects to which no energy is added and from which no energy is removed.

We consider two types of events: explosions and collisions.

Explosions

In an explosion, two (or more) objects start off touching each other and at rest.



The explosion occurs, sending each object off in opposite directions. Because they are travelling in opposite directions, their momentums can add up to 0. This means that momentum will be conserved as momentum before the explosion was 0.

The object with a lower mass moves with a higher velocity, while the object with a higher mass moves with a lower velocity. (This is because of inertia and F=ma.)

The momentum of the two objects is equal but opposite. Because momentum is a vector quantity, the momentums of the two objects add up to give 0. Therefore momentum is conserved – it is the same before and after the explosion.

We say "in a closed system", because if energy was added to or taken away from the two objects, the velocity would be increased or decreased and momentum would not be conserved.

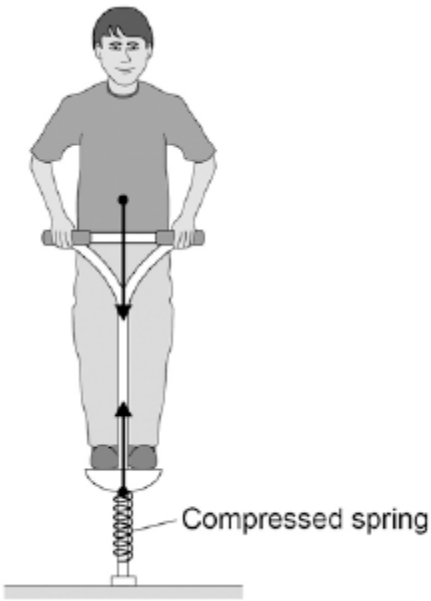
Questions

1. Write out the principle of conservation of momentum
2. What is a closed system?
3. In an explosion, the two pieces always move in opposite directions. Why?
4. Copy the diagram of the bullet and gun and explain how momentum is conserved.

Combined questions

1. The figure below shows the forces acting on a child who is balancing on a pogo stick.

The child and pogo stick are not moving.



(a) The downward force of the child on the spring is equal to the upward force of the spring on the child.

This is an example of which one of Newton's Laws of motion? **(1)**

(b) What energy store is increased in the spring because it is compressed?

(c) The child has a weight of 343 N.

Gravitational field strength = 9.8 N / kg

Calculate the mass of the child.

(e) The weight of the child causes the spring to compress elastically from a length of 30cm to a new length of 23cm.

Calculate the spring constant of the spring.

Give your answer in newtons per metre.

2.

(a) The arrows in the diagram represent the size and direction of the forces on a space shuttle, fuel tank and booster rockets one second after launch. The longer the arrow the bigger the force.

Thrust force

Weight of shuttle, fuel tanks and
booster rockets plus air resistance

(i) Describe the upward motion of the space shuttle one second after launch.

(ii) By the time it moves out of the Earth's atmosphere, the total weight of the space shuttle, fuel tank and booster rockets has decreased and so has the air resistance.

How does this change the motion of the space shuttle?
(Assume the thrust force does not change).

(b) The space shuttle takes 9 minutes to reach its orbital velocity of 8100 m/s.

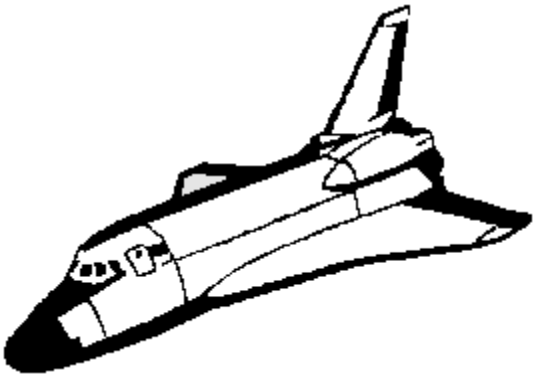
(i) Write down the equation that links acceleration, change in velocity and time taken.

(ii) Calculate, in m/s^2, the average acceleration of the space shuttle during the first 9 minutes of its flight.

(iii) How is the velocity of an object different from the speed of an object?

3

The diagram shows an orbiter, the reusable part of a space shuttle. The data refers to a typical flight.



Orbiter data	
Mass	78 000 kg
Orbital speed	7.5 km/s
Orbital altitude	200 km
Landing speed	100 m/s
Flight time	7 days

(a) (i) What name is given to the force which keeps the orbiter in orbit around the Earth?

(ii) calculate the kinetic energy, in joules, of the orbiter while it is in orbit.

(iii) What happens to most of this kinetic energy as the orbiter re-enters the Earth's atmosphere?

(b) After touchdown the orbiter decelerates uniformly coming to a halt in 50 s.

(i)

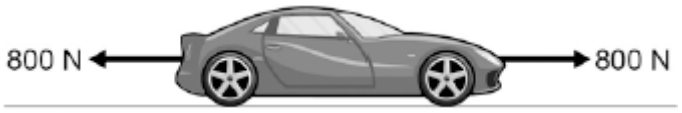
Calculate the deceleration of the orbiter. Show clearly how you work out your answer and give the unit.

(c)

Calculate, in newtons, the force needed to bring the orbiter to a halt. Show clearly how you work out your answer.

4.

The figure below shows the horizontal forces acting on a car.



(1)

a) Will the car be accelerating, decelerating, or travelling forward at a constant speed?

b) During part of the journey the car is driven at a constant speed for five minutes. In this time it covers a distance of 5km. What is the speed of the car?

(1)

(c) During a different part of the journey the car accelerates from 9 m/s to 18 m/s in 6 s.

Calculate the acceleration of the car.

(d) The mass of the car is 1120 kg. The mass of the driver is 80 kg.

Calculate the resultant force acting on the car and driver while accelerating.

(e) Calculate the distance travelled while the car is accelerating.

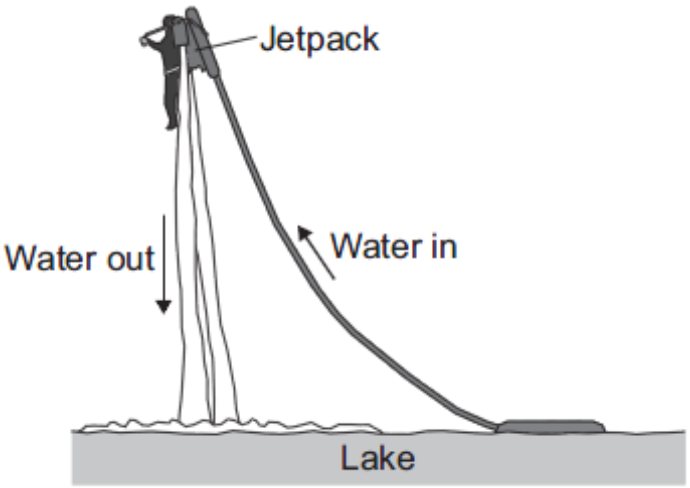
(h) A car driver sees a fallen tree lying across the road ahead and makes an emergency stop.

The braking distance of the car depends on the speed of the car.

For the same braking force, explain what happens to the braking distance if the speed doubles.

You should refer to kinetic energy in your answer.

5. The diagram below shows a person using a device called a jetpack. Water is forced downwards from the jetpack and produces an upward force on the person.



(a) State the condition necessary for the person to be able to remain stationary in mid-air.

(b) The person weighs 700 N and the jetpack weighs 140 N.

(i) Calculate the combined mass of the person and the jetpack.

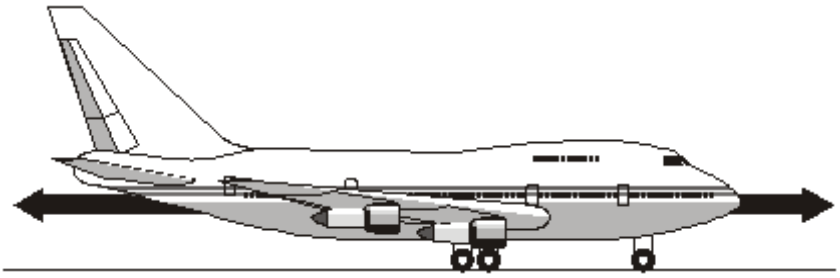
Gravitational field strength = 10 N/kg

(ii) Increasing the upward force to 1850 N causes the person to accelerate upwards.

Calculate the acceleration of the person and the jetpack.

6.

(a) The diagram shows an aircraft and the horizontal forces acting on it as it moves along a runway. The *resultant force* on the aircraft is zero.



(i) What is meant by the term *resultant force*?

(ii) Describe the movement of the aircraft when the resultant force is zero.

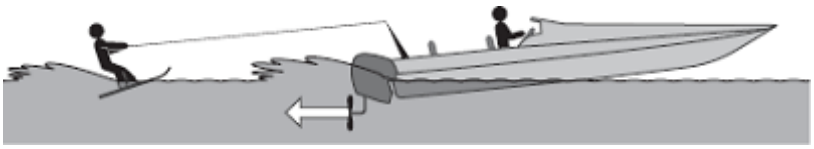
(b) The aircraft has a take-off mass of 320 000 kg. Each of the 4 engines can produce a maximum force of 240 kN.

Calculate the maximum acceleration of the aircraft.

(c) As the aircraft moves along the runway to take off, its acceleration decreases even though the force from the engines is constant.

Explain why.

1. The diagram shows a boat pulling a water skier.



(a) The arrow represents the force on the water produced by the engine propeller.
This force causes the boat to move.

Explain why.

(b) The boat accelerates at a constant rate in a straight line. This causes the velocity of the water skier to increase from 4.0 m/s to 16.0 m/s in 8.0 seconds.

(i) Calculate the acceleration of the water skier and give the unit.

(ii) The water skier has a mass of 68 kg.

Calculate the resultant force acting on the water skier while accelerating.

(iii) How will the force from the boat on the water skier compare to the resultant force on the water skier? Explain your answer.